Unit-2 Combinational Circuit

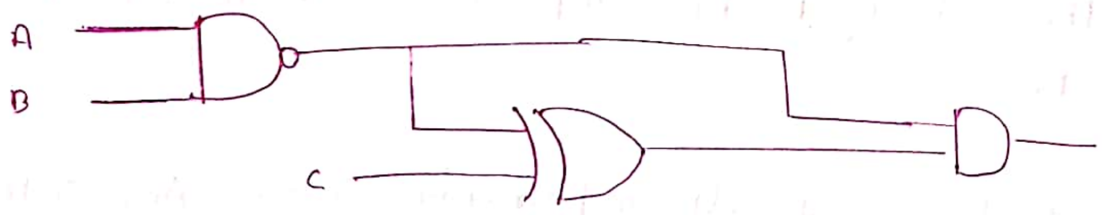
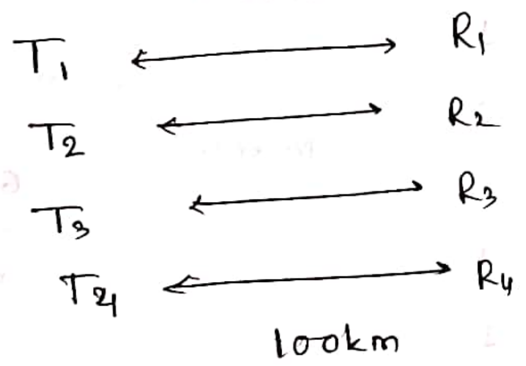


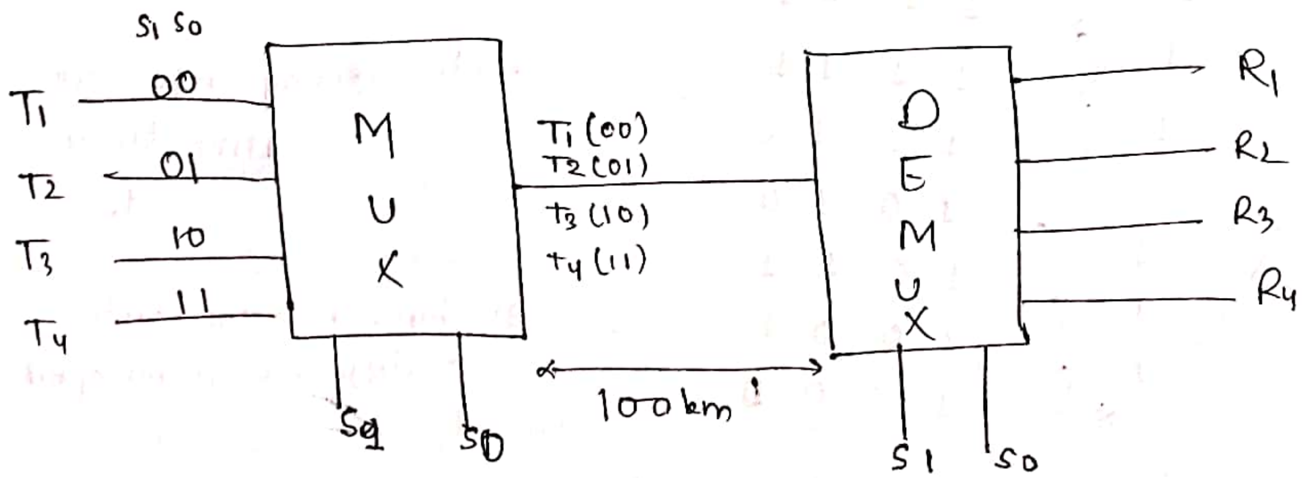
fig. Combinational ckt

· Combinational Ckt is a Circuit made up of Combination of Logic gate. In Combinational ckt the o/p at any time depend on present i/p

* Multiplexer



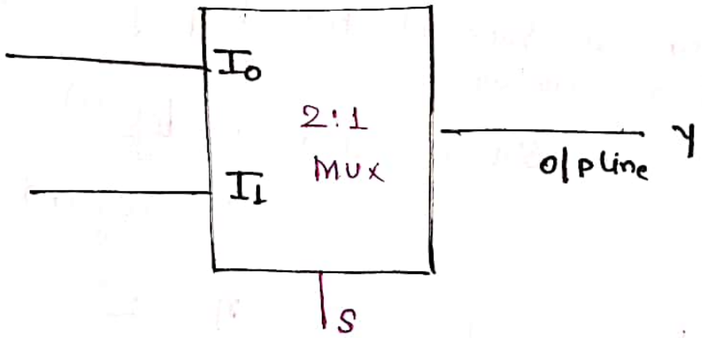
Total Cable required for 1 to 1 X mission = 400km

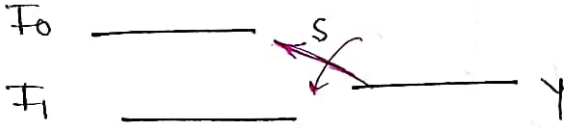


Only 100km Cable Req.

1) 2x1 Mux (2:1 Mux)

2 line to 1 line Mux





S	Y
0	I_0
1	I_1

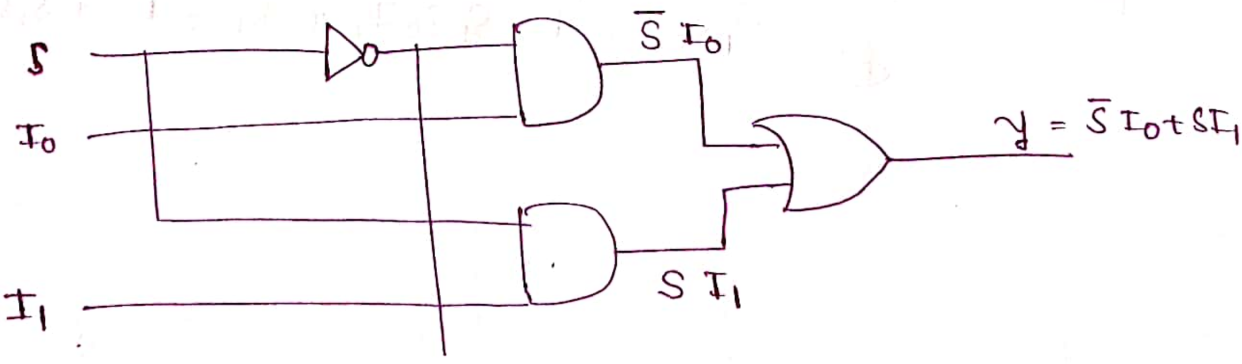
$$Y = \bar{S} I_0 + S I_1$$

Check:- $S = 0$

$$Y = 1 \cdot I_0 + 0 I_1$$
$$Y = I_0$$

$S = 1$

$$Y = 0 I_0 + 1 I_1$$
$$= I_1$$



logic ckt of 2:1 Mux

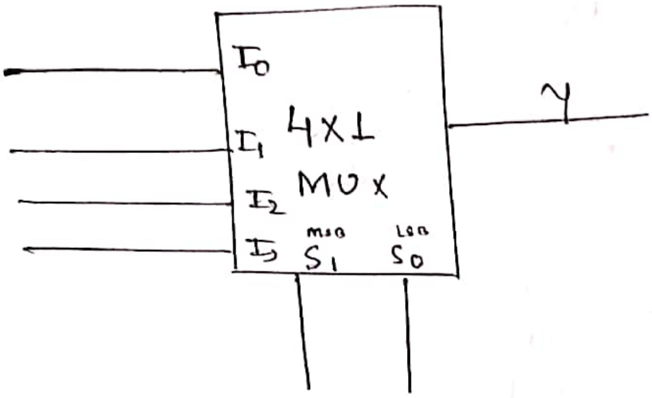
Multiplexer :- It is a Combinational ckt which accepts data through multiple i/p line and passes o/p through Single output line. The Routing of Multiple i/p data into Single o/p line is done with the help of extra i/p line called as Selection line.

Selection line $S = \log_2(n)$: n no of i/p

$S = \log_2(n)$

$n = 2^S$

4 X 1 MUX



Table

S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

Equation :-

$$y = \bar{S_1}\bar{S_0}I_0 + \bar{S_1}S_0I_1 + S_1\bar{S_0}I_2 + S_1S_0I_3$$

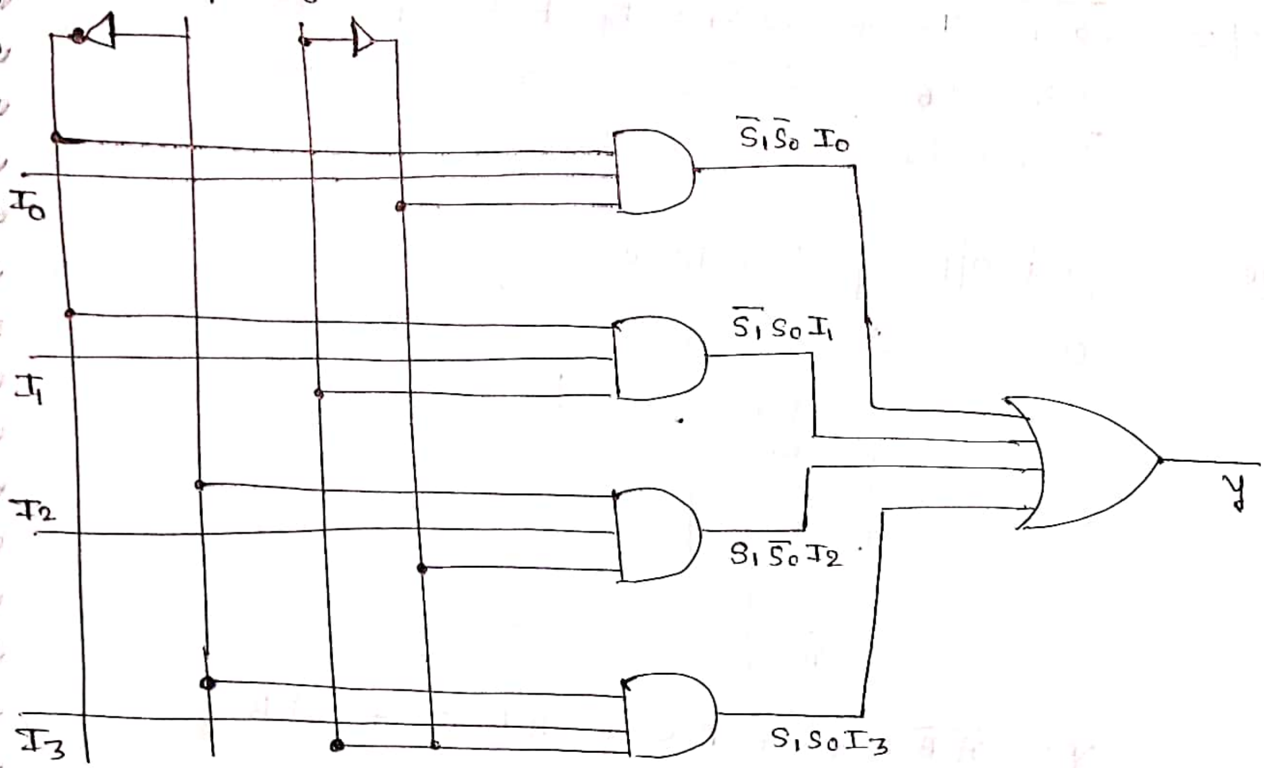
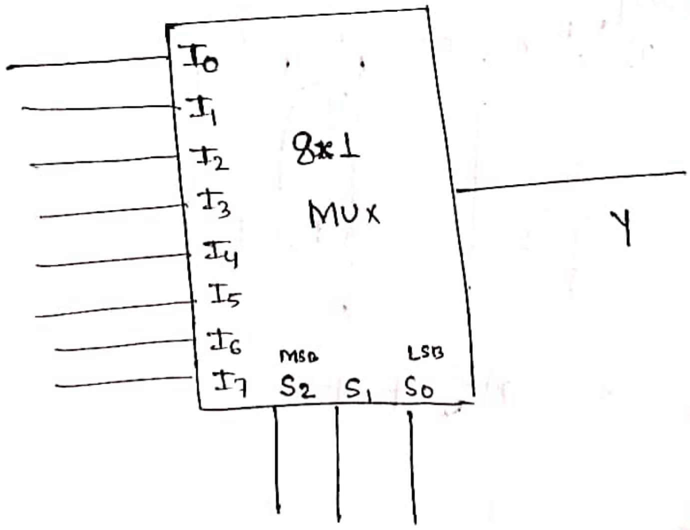


fig. 4x1 MUX logic diagram

8x1 MUX



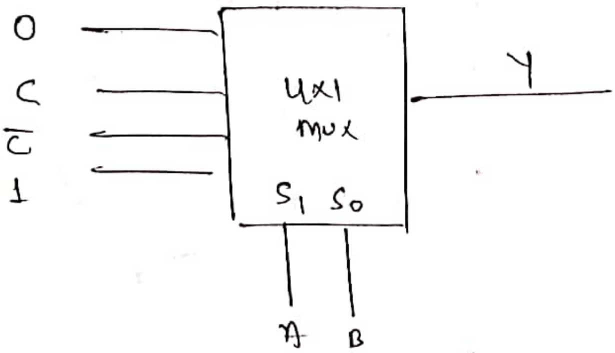
S_0 always LSB

Table

S_2	S_1	S_0	Y
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

$$Y = \overline{S_2 S_1 S_0} I_0 + \bar{S_2}\bar{S_1} S_0 I_1 + \bar{S_2} S_1 \bar{S_0} I_2 + S_2 S_1 \bar{S_0} I_6 + S_2 \bar{S_1} \bar{S_0} I_4 + S_2 \bar{S_1} S_0 I_5 + \bar{S_2} S_1 S_0 I_3 + S_2 S_1 S_0 I_7$$

Que. find o/p of given MUX

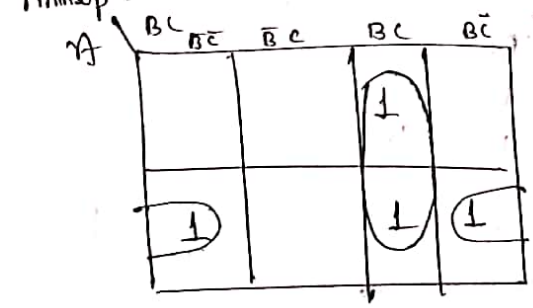


$$Y = \bar{A}\bar{B}\,0 + \bar{A}BC + A\bar{B}\bar{C} + AB$$

$$Y = AB + \bar{A}BC + A\bar{B}\bar{C}$$

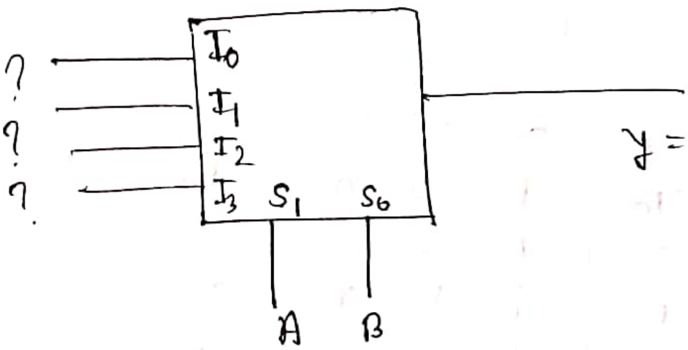
In std SOP $Y_{ssop} = AB\bar{C} + ABC + \bar{A}BC + A\bar{B}\bar{C}$

Y_{minsop} :-



$$Y = A\bar{C} + BC$$

Que.



$$Y = BC + A\bar{C}$$

Sol) $\bar{S}_1 \bar{S}_0 I_0 + \bar{S}_1 S_0 I_1 + S_1 \bar{S}_0 I_2 + S_1 S_0 I_3 = BC + A\bar{C}$

$\bar{A}\bar{B} I_0 + \bar{A} B I_1 + A \bar{B} I_2 + A B I_3 = BC + A\bar{C}$ —— (1)

$Y = BC + A\bar{C}$

$Y = ABC + \bar{A}BC + AB\bar{C} + A\bar{B}\bar{C}$

$Y = ABC + AB\bar{C} + \bar{A}BC + A\bar{B}\bar{C} + \bar{A}\bar{B}\cdot 0$

$Y = AB\cdot 1 + \bar{A}BC + A\bar{B}\bar{C} + \bar{A}\bar{B}\cdot 0$ —— (2)

Compare eq (1) & (2)

$I_3 = 1$

$I_2 = \bar{C}$

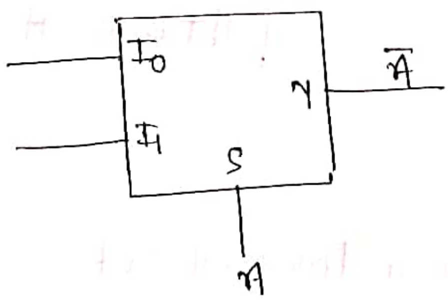
$I_0 = 0$

$I_1 = C$

obtaining o/p is called designing
getting o/p is called Analysis

Que. Design following function using 2:1 MUX

(1) NOT $Y = \bar{A}$



$Y = \bar{S} I_0 + S I_1$

$Y = \bar{A} I_0 + A I_1$

$I_0 = 1$, $I_1 = 0$

$Y = \bar{A}$ //.

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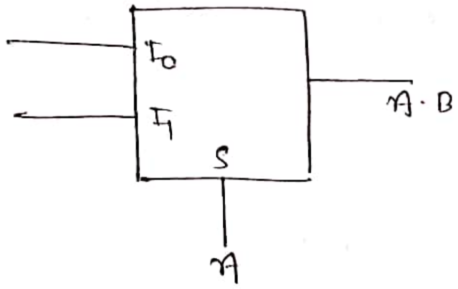
2. AND $Y = A \cdot B$



$Y = \bar{S} I_0 + S I_1$

$Y = \bar{A} I_0 + A I_1$

$Y = AB + \bar{A} \cdot 0$

$Y = AB$

$I_1 = B$

$I_0 = 0$ //.

3. OR $Y = A + B$

$Y = A + B(A + \bar{A})$

$= A + AB + \bar{A}B$

$= A(1+B) + \bar{A}B$

$= A \cdot 1 + \bar{A}B$

$Y = I_0 \bar{A} + I_1 A$

$\cancel{Y = I_0 \bar{A} + I_1 A + A \cdot 0}$

$\cancel{I_0 = I_1 = B}$

$\boxed{A + B = A + \bar{A}B}$

$Y = I_0 \bar{A} + I_1 A$

on Comp.

$I_0 = B \quad I_1 = 1$

note:- 2 to 1 MUX is also a Universal ckt

4) NOR gate

$$y = \overline{A+B}$$

$$y = \bar{A} \cdot \bar{B}$$

Take: $S = A$

$$y = \bar{S} \cdot I_0 + S I_1$$

$$y = \bar{A} I_0 + A I_1$$

$$y = \bar{A}\bar{B} + A \cdot 0$$

on Comp. $\bar{B} = I_0$, $I_1 = 0$ //

5) $$y = \overline{A \cdot B}$$

$$y = \bar{A} + \bar{B}$$

$$= \bar{A} \cdot 1 + \bar{A}\bar{B} + A\bar{B}$$

$$= \bar{A}(1 + \bar{B}) + A\bar{B}$$

$$= \bar{A} \cdot 1 + A\bar{B}$$

OR

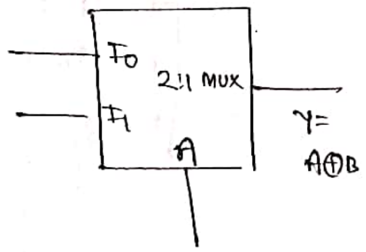
$$y = \bar{A} + \bar{B}$$

$$= \bar{A} + A\bar{B}$$

$I_0 = 1$ $\quad$ $I_1 = \bar{B}$

6) $$y = A \oplus B = \bar{A}B + A\bar{B}$$

$I_0 = B$ $\quad$ $I_1 = \bar{B}$



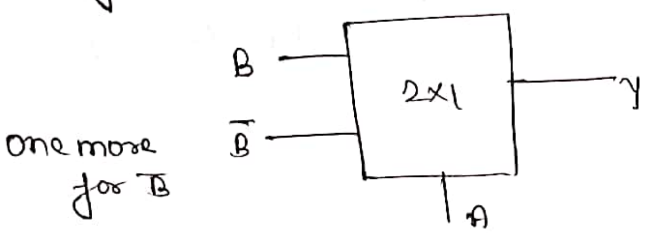
7) $$y = A \odot B = \bar{A}\bar{B} + AB$$

$I_0 = \bar{B}$ $\quad$ $I_1 = B$

Que 1 What are the Minimum number of 2:1 MUX required to generate a two i/p AND gate and a two input EXOR gate

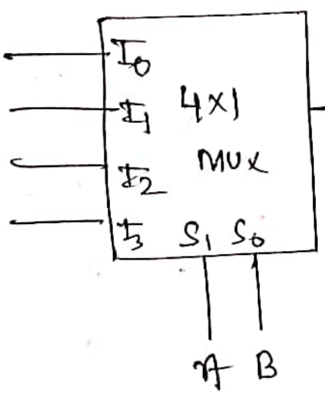
Sol for two i/p AND gate
Only 1 2x1 MUX Req

for two i/p EXOR gate = 2 2x1 MUX



Ans. 1 & 2 //.

Que. Design using 4x1 MUX
1) $Y = A \oplus B$
$Y = \bar{A}B + A\bar{B}$



$Y = \bar{S_1}\bar{S_0}I_0 + \bar{S_1}S_0 I_1 + S_1\bar{S_0}I_2 + S_1 S_0 I_3$

$= \bar{A}\bar{B}I_0 + \bar{A}BI_1 + A\bar{B}I_2 + ABI_3$

On Compar.

$I_0 = 0$

$I_3 = 0$

$I_1 = 1$

$I_2 = 1$

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II) $y = \overline{A \cdot B}$

Std. eq^n $\quad Y = \bar{A}\bar{B} I_0 + \bar{A} B I_1 + A \bar{B} I_2 + A B I_3$

$$y = \bar{A} + \bar{B}$$

$$= \bar{A} B + \bar{A}\bar{B} + \bar{A}\bar{B} + A\bar{B}$$

$$= \bar{A}\bar{B} + \bar{A} B + A \bar{B} + A B \cdot 0$$

On Comp.

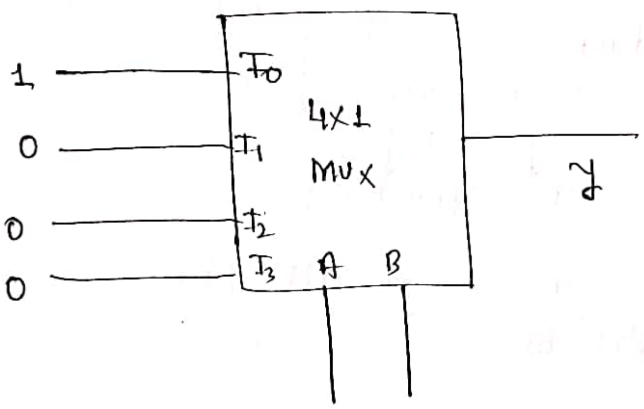
$I_0 = 1$

$I_1 = 1$

$I_2 = 1$

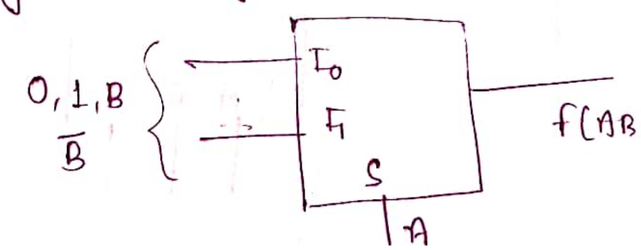
$I_3 = 0$ //

III) $y = \overline{A + B}$



Imp pts.

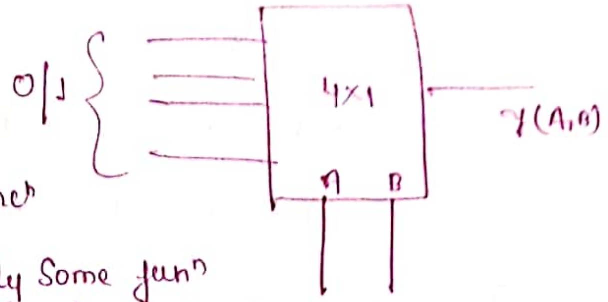
1. With a 2x1 Mux, we can design some but not all functions of 2 variable



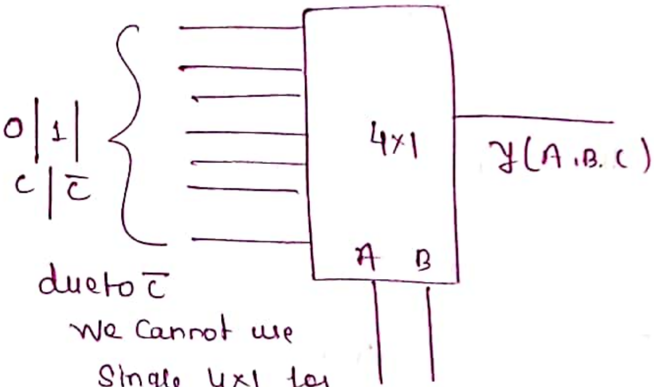
2. With 2x1 MUX and 1 Inverter we can design all the function of two Variable

3. With 4x1 MUX we can design all the function of two Variables

4. With 4x1 Mux we Cannot design all the func^n of three Variable. Only some fun^n Can be designed

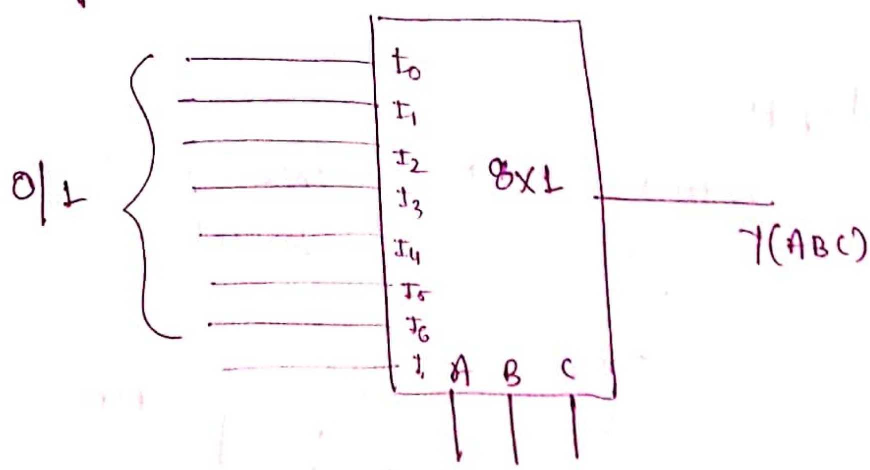


5. With 4x1 MUX and Inverter we can design all the function of three Variable



due to c̄
we cannot use
single 4x1 for
all Y(A,B,C).

6. with 8x1 Mux we can design all the function of 3 Variables



Que. Without any additional Circuitry an 8X1 MUX Can be used to obtain

1) Some but not all Boolean function.

II) All function of three Var. but none of four Variables

• III) All function of three Var and Some but not all of four Variable.

IV) All function of four Variable.

Que. Suppose only one MUX and one Inverter are allowed to be used to Implement any Boolean function of n Variable. what is min. Size of Mux need

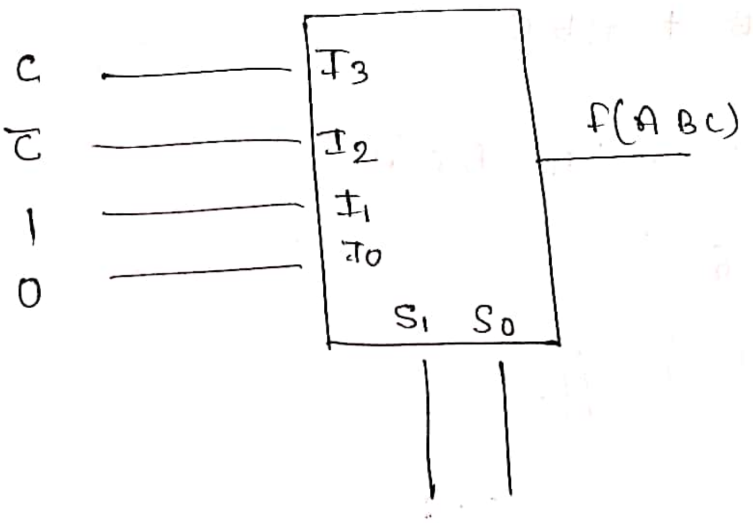
A) 2^n line to 1 line

B) 2^{n+1} line to 1 line

c) 2^{n-1} line to 1 line

d) 2^{n-2} line to 1 line

*Que. Find o/p in SSOP (Minterm no. form).



$y = \overline{S_1}\,\overline{S_0}\,I_0 + \overline{S_1}\,S_0 I_1 + S_1 \overline{S_0}\,I_2 + S_1 S_0 I_3$

from ckt

$y = \cancel{\overline{S_1}\,\overline{S_0}\,I_0}$

$\overline{S_1}\,\overline{S_0}\,0 + \overline{S_1}\,S_0\,1 + S_1\overline{S_0}\,\overline{C} + S_1 S_0 C$

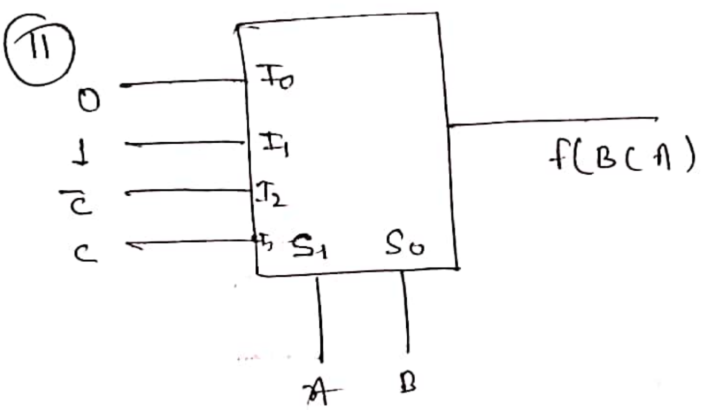
$y = \overline{A}\,\overline{B}\cdot 0 + \overline{A}\,B + A\overline{B}\,\overline{C} + ABC$

$y = \overline{A}B + A\overline{B}\,\overline{C} + ABC$

$\Sigma m(2, 3, 4\ 7)$ //.

{ I_0 will always multiplied by $\overline{S_0}\,\overline{S_1}$ position will not change its value }

(II)



$\Rightarrow \overline{S_1}\,\overline{S_0}\,I_0 + \overline{S_1}\,S_0 I_1 + S_1\overline{S_0}\,I_2 + S_1 S_0 I_3$

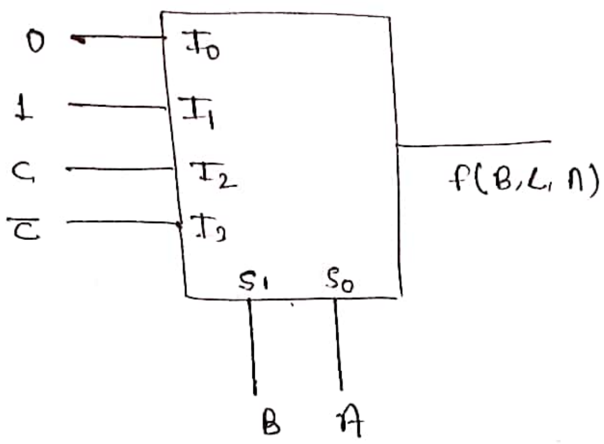
$\Rightarrow A\overline{B}\,\overline{C} + \overline{A}B + ABC$

$\Sigma m(BCA,\ \overline{B}\,\overline{C}A,\ B\overline{C}\,\overline{A},\ BC\overline{A})$

$\Sigma m(7, 1, 4, 6)$

$= \Sigma m(1\ 4\ 6\ 7)$ //.

③



Sol

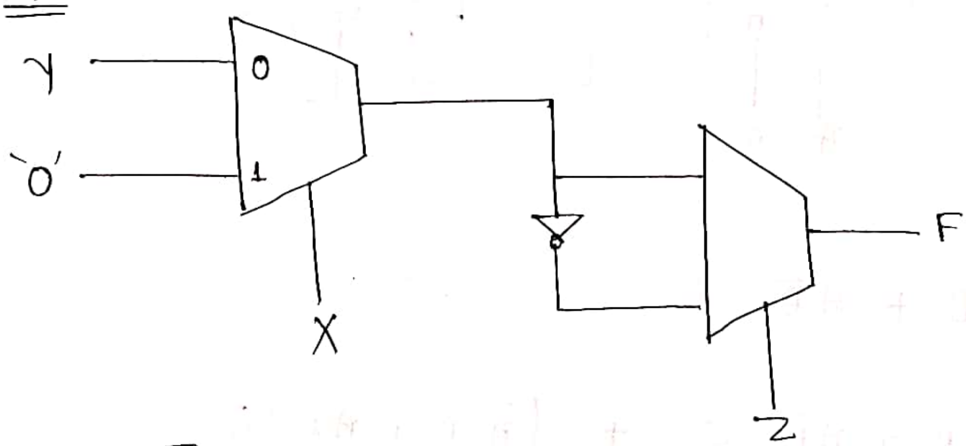
$$y = \bar{B}\bar{A}\cdot 0 + \bar{B}A + B\bar{A}C + BA\bar{C}$$

$$= \bar{B}A + B\bar{A}C + BA\bar{C}$$

$$= \bar{B}CA + \bar{B}\cdot\bar{C}A + B\bar{A}C + BA\bar{C}$$

$$\Sigma m(3,1,5,6) = \Sigma m(1,3\ 5\ 6) \quad //$$

Q4.



A) $\bar{X}\bar{Y}\bar{Z} + XY + \bar{Y}Z$

B) $\bar{X}Y\bar{Z}$

Sol

$$Y' = \bar{X}y + X \cdot 0$$

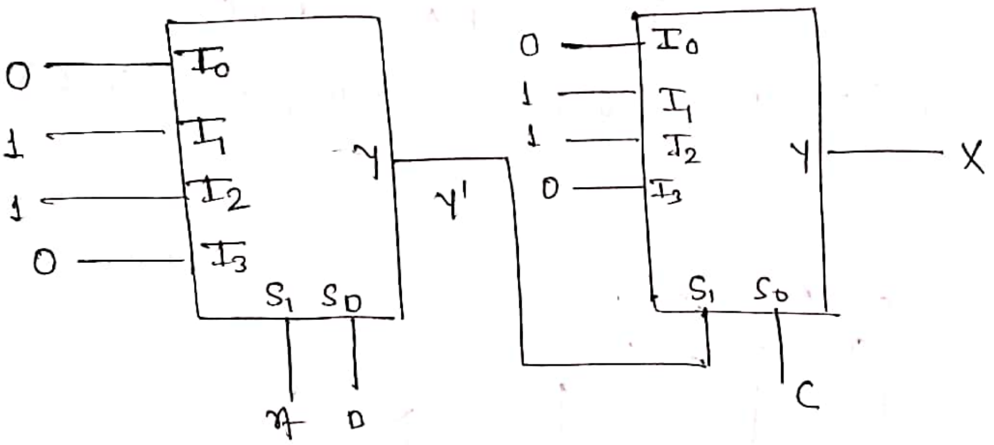
$$Y' = \bar{X}Y$$

$$\overline{\bar{X}Y} = X + \bar{Y}$$

$$Y = \bar{Z}\,\bar{X}Y + Z(X+\bar{Y})$$

$$= \bar{X}Y\bar{Z} + XZ + Z\bar{Y} \quad //$$

Que2



$$Y' = \bar{A}B + A\bar{B}$$

$$X = \overline{\bar{A}B + A\bar{B}}\,C + (\bar{A}B + A\bar{B})\bar{C}$$

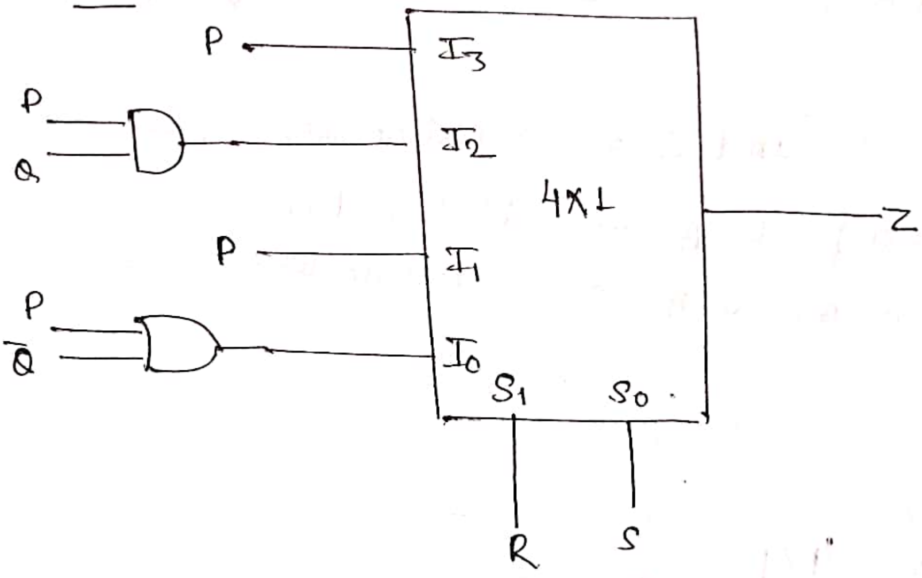
$$\Rightarrow (\overline{\bar{A}B} \cdot \overline{A\bar{B}})C + \bar{A}B\bar{C} + A\bar{B}\bar{C}$$

$$\Rightarrow (A+\bar{B} \cdot \bar{A}+B)C + \bar{A}B\bar{C} + A\bar{B}\bar{C}$$

$$\Rightarrow ABC + \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} \quad //$$

$\Rightarrow$

Que.



find Z

Sol

$$Z = \bar{R}\bar{S}(P+\bar{Q}) + \bar{R}SP + R\bar{S}(PQ) + RSP$$

$$= PQ\bar{R}\bar{S} + P\bar{Q}\bar{R}\bar{S} + \bar{P}\bar{Q}\bar{R}\bar{S} + P\bar{Q}\bar{R}\bar{S}$$

$$+ P\bar{Q}\bar{R}S + PQ\bar{R}S + PQR\bar{S} + P\bar{Q}RS + PQRS$$

PQ \ RS	$\bar{R}\bar{S}$	$\bar{R}S$	RS	$R\bar{S}$
$\bar{P}\bar{Q}$	1			
$\bar{P}Q$				
PQ	1	1	1	1
$P\bar{Q}$	1	1	1	

$$Z = PQ + PS + \bar{Q}\bar{R}\bar{S}$$

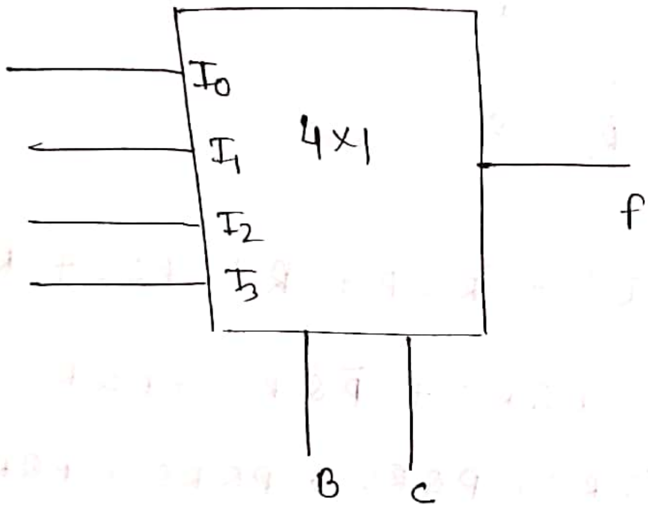
Que. Design $f(A,B,C) = \sum m(1,2,4,6,7)$ using 4x1 MUX

i) Using B (MSB) and C as Selection line

ii) Using A (MSB) & B as Selection line

iii) Using C (MSB) & A as Selection line

Sol i)



$f(A,B,C) =$

$\bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + AB\bar{C} + ABC$ — (1)

$F = \bar{B}\bar{C} I_0 + \bar{B}C I_1 + B\bar{C} I_2 + BC I_3$ — (2)

$= I_1 \bar{B}C + I_2 B\bar{C} + I_0 \bar{B}\bar{C} + I_3 BC$

on comparing (1) & (2)

$I_0 = A$

$I_1 = \bar{A}$

$I_2 = \cancel{A}\ 1$

$I_3 = A$ //

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By Tabulation Method

$I's \Leftarrow$; Selection line Variable $\Rightarrow$

	I_0 (00) $\bar{B}\bar{C}$	I_1 (01) $\bar{B}C$	I_2 (10) $B\bar{C}$	I_3 (11) BC
$A=0$	0	1	2	3
$A=1$	4	5	6	7
	A	$\bar{A}$	$A+\bar{A}=1$	A

$I_0 = A$

$I_1 = \bar{A}$

$I_2 = 1$

$I_3 = A$ //

ii)

	$\bar{A}\bar{B}$ I_0 (00)	$\bar{A}B$ I_1 (01)	$A\bar{B}$ I_2 (10)	AB I_3 (11)
$C=0$	0	2	4	6
$C=1$	1	3	5	7
	C	$\bar{C}$	$\bar{C}$	1

iii)

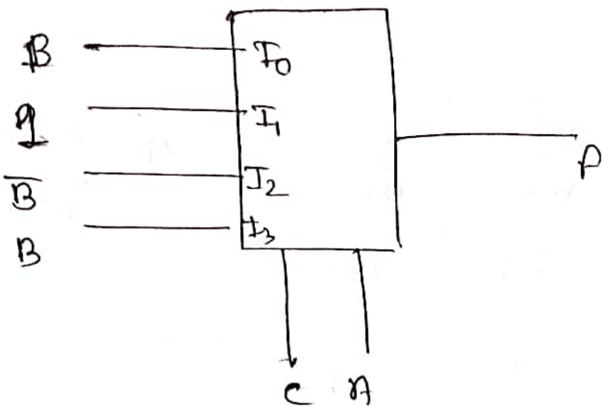
	$\bar{C}\bar{A}$ I_0 (00)	$\bar{C}A$ I_1 (01)	$C\bar{A}$ I_2 (10)	CA I_3 (11)
$\bar{B}$ 0	0	4	1	5
B 1	2	6	3	7
	B	$\bar{B}$ 1	$\bar{B}$	B

$I_0 = B$

$I_1 = 1$

$I_2 = \bar{B}$

$I_3 = B$



Que. $F(A,B,C,D) = \sum m(0,3,4,5,7,8,10,12,14,15)$

i) Using 8x1 MUX and

B (MSB) C & D (LSB) as Selection line

ii) A (MSB), C & B (LSB) as Selection line

Sol

i)

	000 I_0 $\bar B\bar C\bar D$	001 I_1 $\bar B\bar C D$	010 I_2 $\bar B C\bar D$	011 I_3 $\bar B C D$	100 I_4 $B\bar C\bar D$	101 I_5 $B\bar C D$	110 I_6 $B C\bar D$	111 I_7 BCD
A=0	0	1	2	3	4	5	6	7
A=1	8	9	10	11	12	13	14	15
	1	0	A	$\bar A$	1	$\bar A$	A	1
	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7

ii)

	000 I_0 $\bar A\bar C\bar B$	001 I_1 $\bar A\bar C B$	010 I_2 $\bar A C\bar B$	011 I_3 $\bar A C B$	100 I_4 $A\bar C\bar B$	101 I_5 $A\bar C B$	110 I_6 $A C\bar B$	111 I_7 ACB
D=0	0	4	2	6	8	12	10	14
D=1	1	5	3	7	9	13	11	15
	$\bar D$	1	D	D	$\bar D$	$\bar D$	$\bar D$	1

+1

D is LSB
So second Row is +1 of first

III) Using 4x1 MUX & C(MSB) & D(LSB) of Selection line

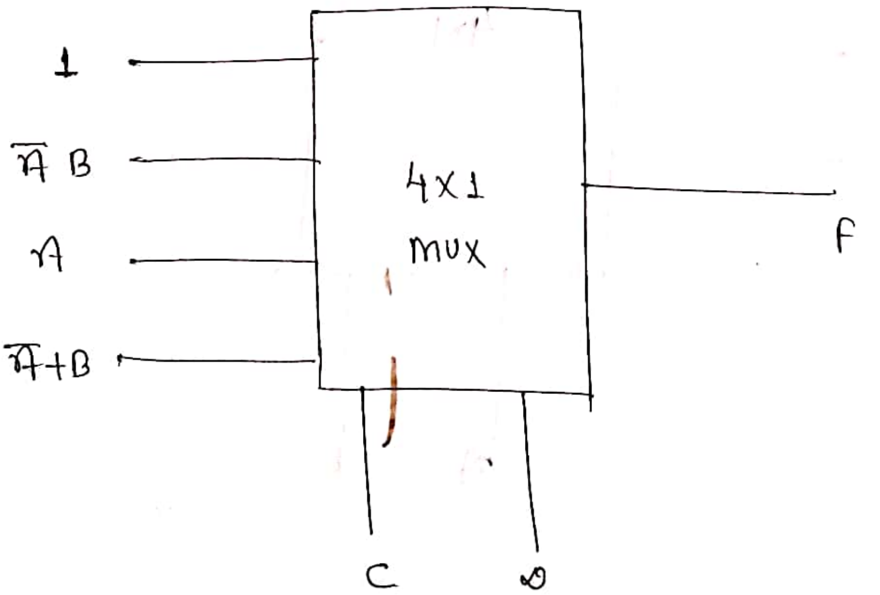
Sol

	I_0 00 $\bar{C}\bar{D}$	I_1 01 $\bar{C}D$	I_2 10 $C\bar{D}$	I_3 11 CD
$\bar{A}\bar{B}$ 00	0	1	2	3
$\bar{A}B$ 01	4	5	6	7
$A\bar{B}$ 10	8	9	10	11
AB 11	12	13	14	15
	1	$\bar{A}B$	A	$\bar{A}+AB$

$\bar{A}+AB$

$(\bar{A}+A)(\bar{A}+B)$

$= \bar{A}+B$

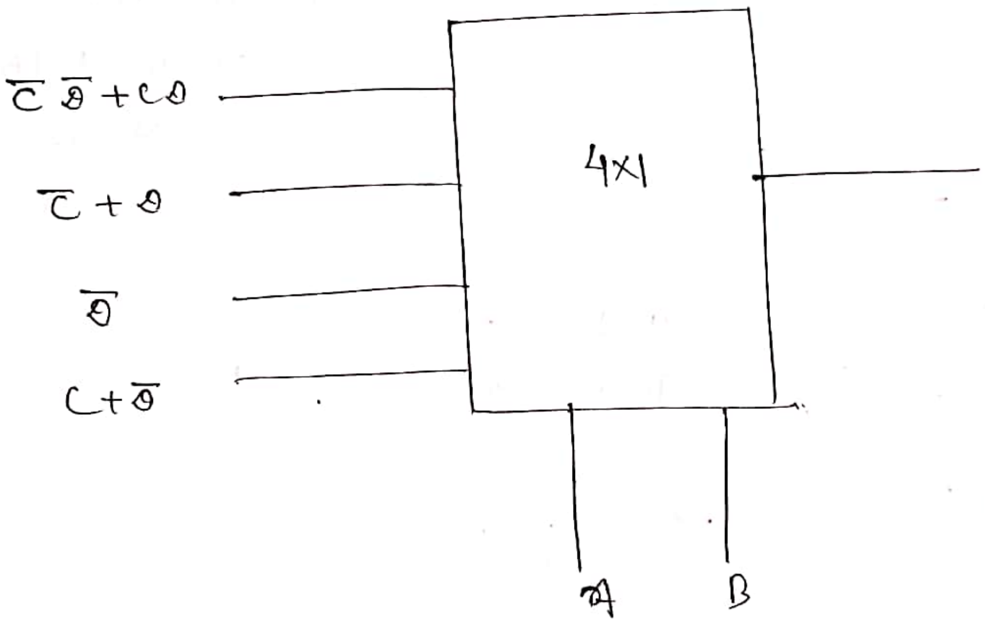


IV) Using 4X1 MUX & A(MSB) & B(LSB) of Selection line

		$\overline{A}\,\overline{B}$ 00	$\overline{A}B$ 01	$A\overline{B}$ 10	AB 11
$\overline{C}\,\overline{D}$	00	0	4	8	12
$\overline{C}D$	01	1	5	9	13
$C\overline{D}$	10	2	6	10	14
CD	11	3	7	11	15

chose this no. as Max term i.e $C+\overline{D}$

I_0	I_1	I_2	I_3
$\overline{C}\,\overline{D}+CD$	$\overline{C}+D$	$\overline{D}$	$C+\overline{D}$



v) Using (2x1) MUX and A as Selection line

	I_0 $\bar{A}$ 0	I_1 A 1
$\bar{B}\bar{C}\bar{D}$	0	8
$\bar{B}\bar{C}D$	1	9
$\bar{B}C\bar{D}$	2 3	10 11
$B\bar{C}\bar{D}$	4	12
$B\bar{C}D$	5	13
$BC\bar{D}$	6	14
$\bar{B}\bar{C}\bar{D}$	7	15
BCD		

~~$(B+C+D)$~~
~~$(B+\bar{C}+D)$~~
~~$(\bar{B}+\bar{C}+D)$~~

Using kmap :- Sop

B \ CD	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{B}$	1	0	0	1
B	1	0	1	1

$I_1 = \bar{D} + BC$

B \ CD	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{B}$	1			1
B	1		1	1

$I_0 = \bar{C}\bar{D} + C\bar{D} + BD$

Que. A Boolean function $f(A,B,C,D) = \prod M(1, 5, 12\ 15)$ is to be Implemented using 8×1 MUX (A is MSB) Inputs $A\,B\,C$ are Connected to Select Inputs $S_2\ S_1\ S_0$ of MUX resp. Which one of the following option are Correct Input, to pins 0 1 2 3 4 5 6 7 in order?

A) D 0 D 0 0 0 $\bar D$ $\bar D$

B) $\bar D$ 1 $\bar D$ 1 1 1 D $\bar D$

Sol $f(ABCD) = \prod M(1, 5, 12\ 15)$

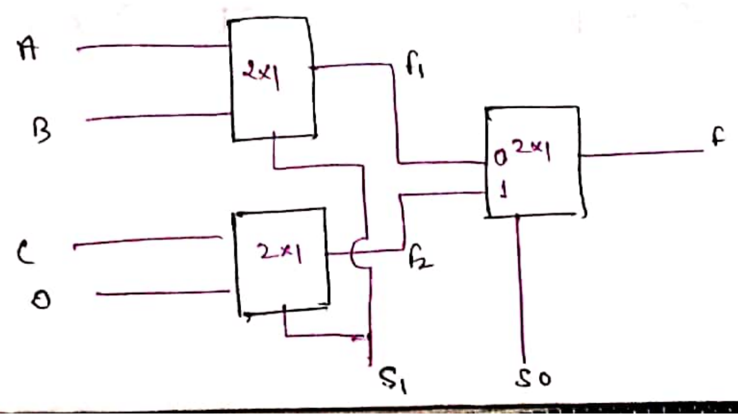
~~A+B+C~~ ~~A+B+C~~

	$\bar A\bar B\bar C$ 000	$\bar A\bar B C$ 001	$\bar A B\bar C$ 010	$\bar A B C$ 011	$A\bar B\bar C$ 100	$A\bar B C$ 101	$AB\bar C$ 110	ABC 111
$D=0$	0	2	4	6	8	10	12	14
$D=1$	1	3	5	7	9	11	13	15
	$\bar D$	1	$\bar D$	1	1	1	D	$\bar D$

(Mark all Minterm)

<u>opt B</u> //.

Que. Define the operation of given ckt



Sol

$f_1 = \overline{S_1} A + S_1 B$

$f_2 = \overline{S_1} C + S_1 D$

$f = \overline{S_0} F_1 + S_0 f_2$

$= \overline{S_0}(\overline{S_1} A + S_1 B) + S_0(\overline{S_1} C + S_1 D)$

$= \overline{S_0}\,\overline{S_1} A + \overline{S_0} S_1 B + S_0 \overline{S_1} C + S_0 S_1 D$

$= \overline{S_1}\,\overline{S_0} A + S_1 \overline{S_0} B + \overline{S_1} S_0 C + S_1 S_0 D$

S_1	S_0	f
0	0	A
0	1	C
1	0	B
1	1	D

if $S_0 \longleftrightarrow S_1$ o/p MUX MSB

then

S_0	S_1	f
0	0	A
0	1	B
1	0	C
1	1	D

Conclusion :- Most. Significant bit of Selection line is assinged to MUX which produce Output

Que. A Boolean function $f(A,B,C,D) = \Pi M(1,5,12\ 15)$ is to be Implemented using 8×1 MUX (A is MSB) Inputs A B C are Connected to Select Inputs $S_2\ S_1\ S_0$ of MUX resp. which one of the following option are Correct Input, to pins 0 1 2 3 4 5 6, 7 in order ?

	0	1	2	3	4	5	6	7
A)	D	0	D	0	0	0	$\bar D$	$\bar D$
B)	$\bar D$	1	$\bar D$	1	1	1	D	$\bar D$

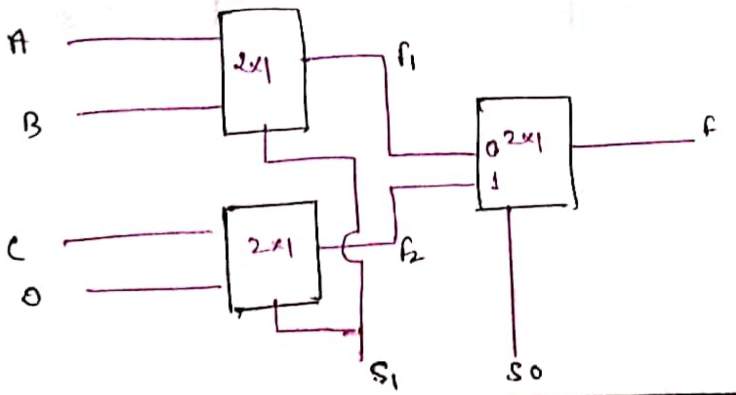
Sol $f(ABCD) = \Pi M(1, 5, 12\ 15)$

	$\bar A\bar B\bar C$ 000	~~A+B+C~~ $\bar A\bar B C$ 001	~~A+B+C~~ $\bar A B\bar C$ 010	$\bar A B C$ 011	$A\bar B\bar C$ 100	$A\bar B C$ 101	$AB\bar C$ 110	ABC 111
$D=0$	0	2	4	6	8	10	12	14
$D=1$	1	3	5	7	9	11	13	15
	$\bar D$	1	$\bar D$	1	1	1	D	$\bar D$

(Mark all Minterm)

opt B //.

Que. Define the operation of given ckt



Sol

$f_1 = \overline{S_1} A + S_1 B$

$f_2 = \overline{S_1} C + S_1 D$

$f = \overline{S_0} f_1 + S_0 f_2$

$= \overline{S_0} \left(\overline{S_1} A + S_1 B \right) + S_0 \left(\overline{S_1} C + S_1 D \right)$

$= \overline{S_0}\, \overline{S_1} A + \overline{S_0} S_1 B + S_0 \overline{S_1} C + S_0 S_1 D$

$= \overline{S_1}\, \overline{S_0} A + S_1 \overline{S_0} B + \overline{S_1} S_0 C + S_1 S_0 D$

S_1	S_0	f
0	0	A
0	1	C
1	0	B
1	1	D

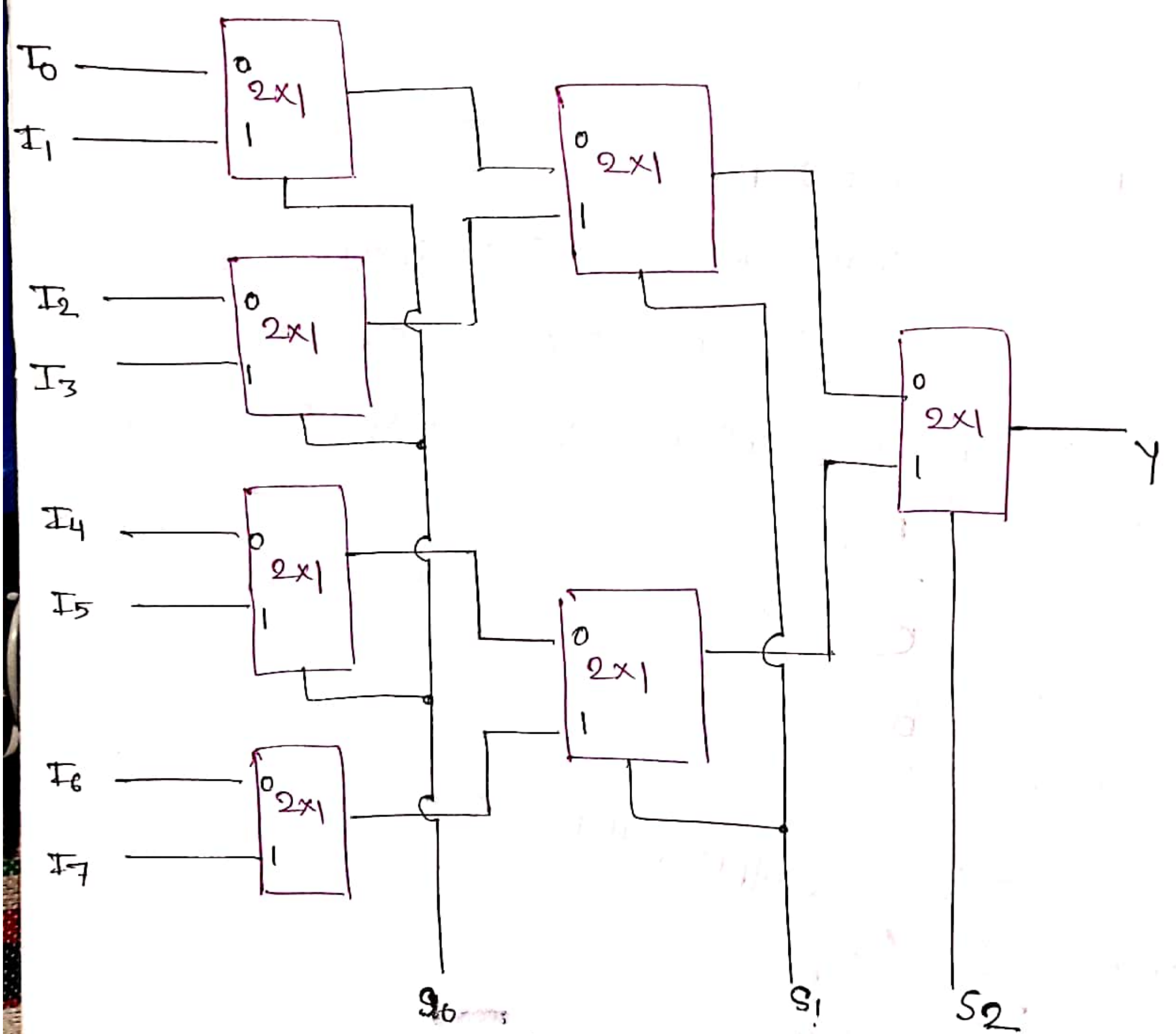
if $S_0 \longleftrightarrow S_1$ o/p MUX MSB

then

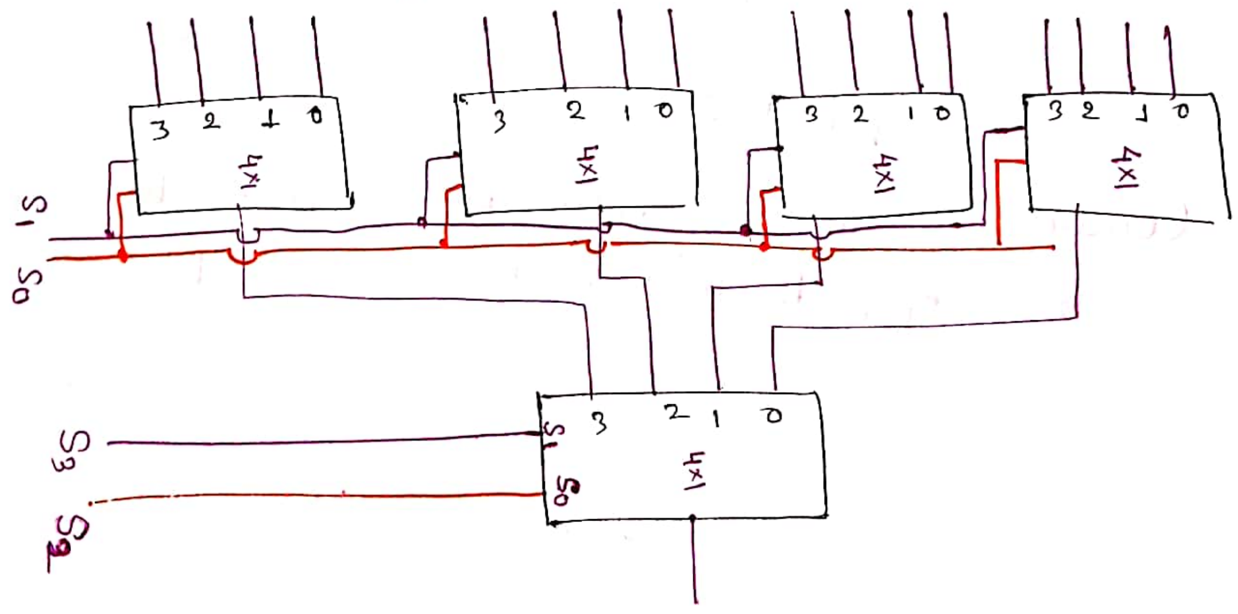
S_0	S_1	f
0	0	A
0	1	B
1	0	C
1	1	D

Conclusion :- Most Significant bit of Selection line is assigned to MUX which produce Output

* 8 x 1 MUX using 2x1 MUX



* 16 x 1 MUX using 4x1



$S_3 S_2 S_1 S_0$ Wrong

$S_3 S_2$ $S_1 S_0$ Correct (each have 4 ~~value~~ terms)

Note:-

number of MUX Required

(I) 4x1 using 2x1

$\frac{4}{2}$ $\frac{2}{2}$

2 + 1 = 3
Level 1, Level 2 (O/P)

(II) 8x1 using 2x1

$\frac{8}{2}$ $\frac{4}{2}$ $\frac{2}{2}$

4 2 1 = 7

(III) 16x1 using 4x1

$\frac{16}{4}$ $\frac{4}{4}$

4 1 = 5

(IV) 64x1 using 4x1

$\frac{64}{4}$ $\frac{16}{4}$ $\frac{4}{4}$

16 4 1 = 21
Level 1, Level 2, Level 3 (O/P)

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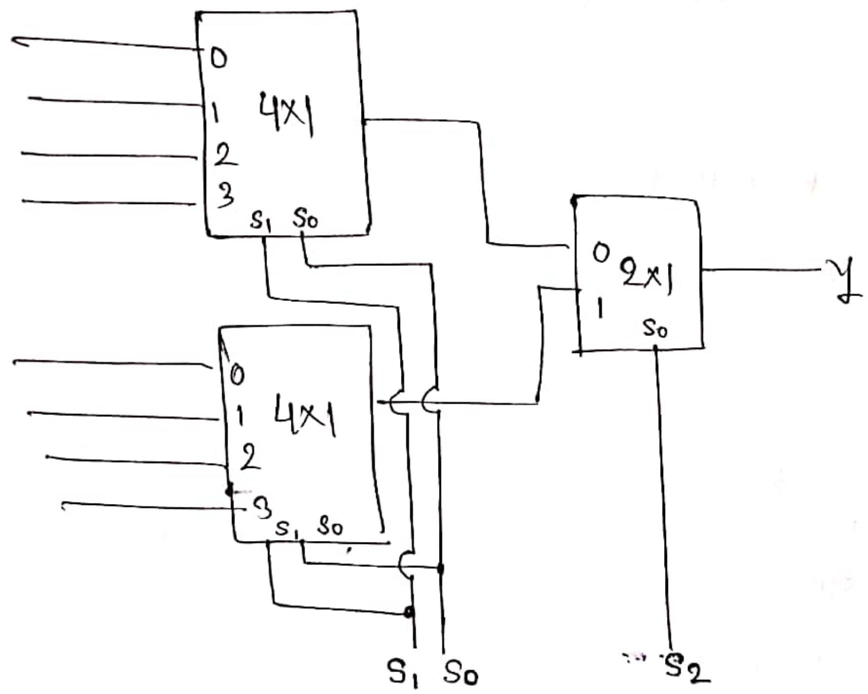
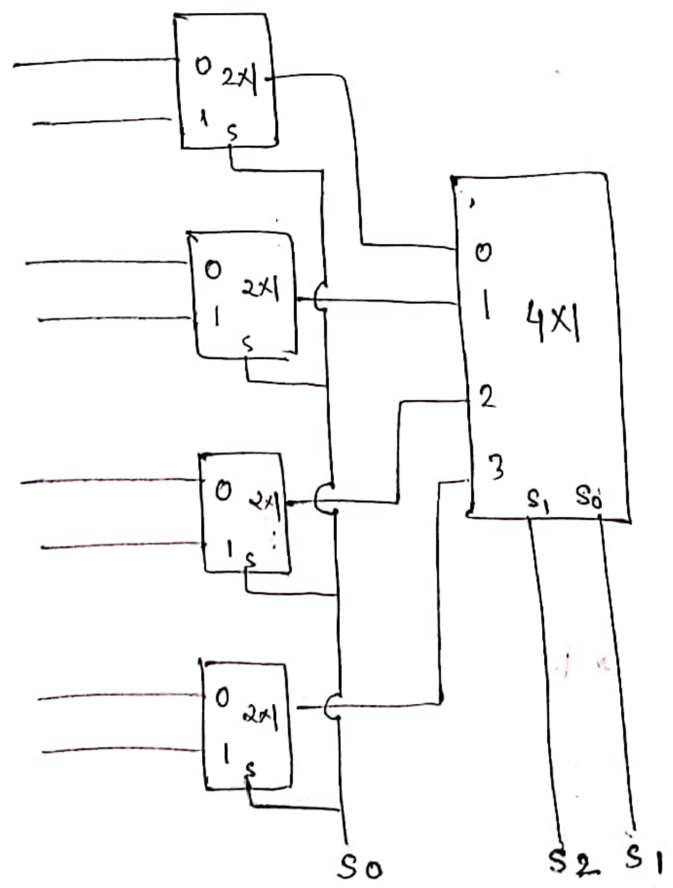
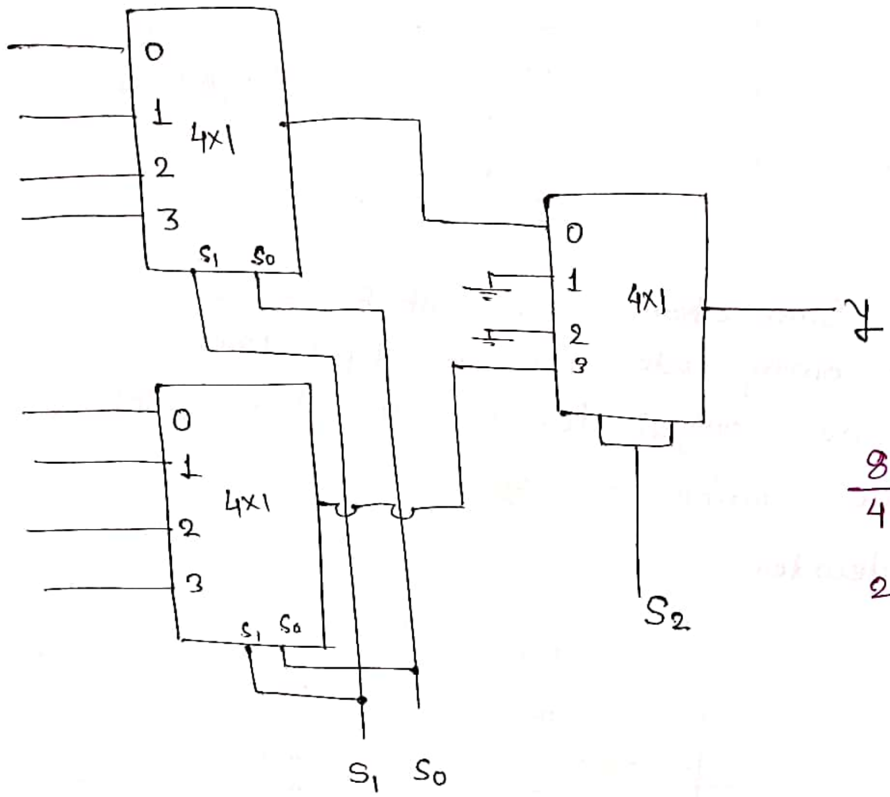
* 8x1 MUX using 4x1 & 2x1



OR



* 8x1 using 4x1

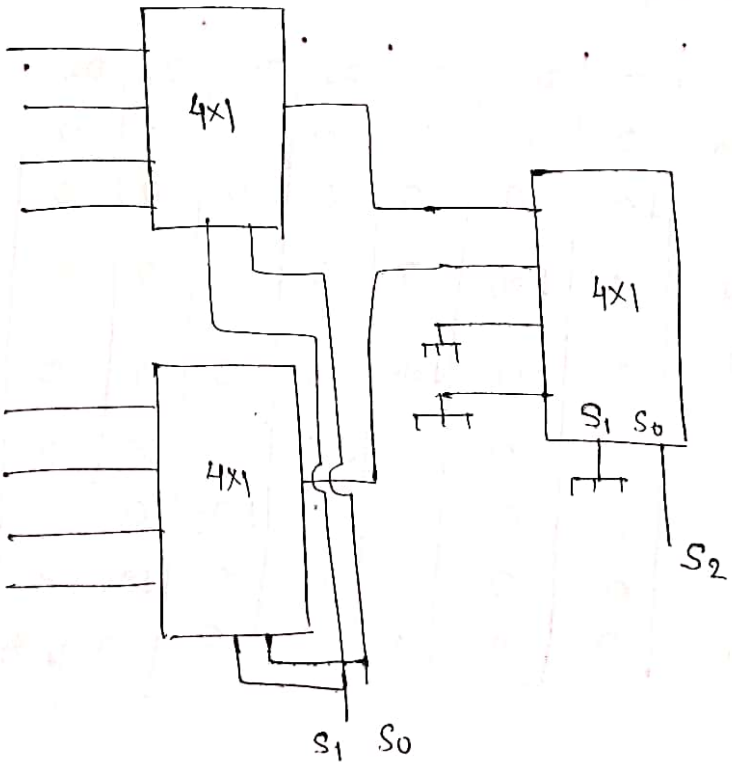


$\frac{8}{4}$ $\frac{2}{4}$

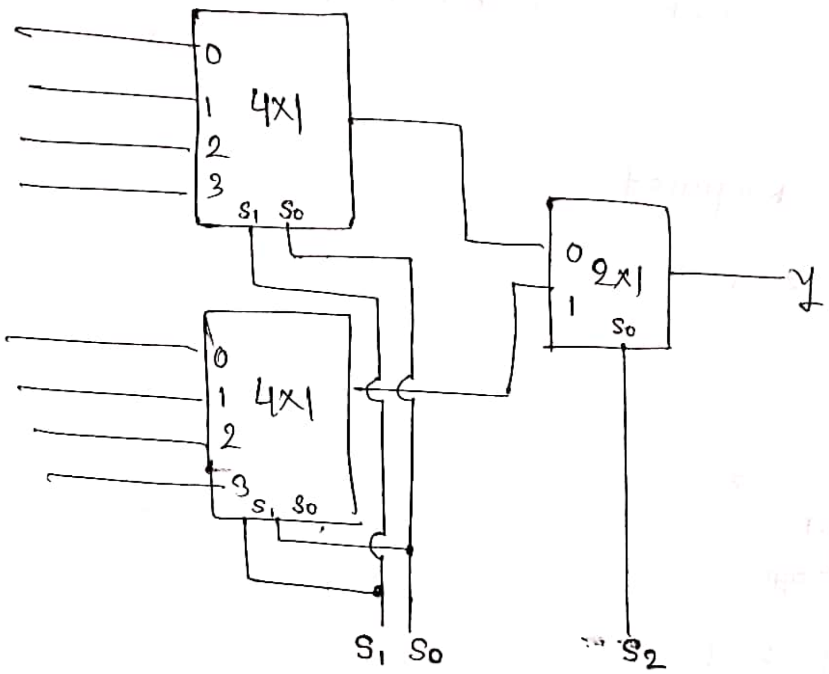
$2 \quad 1 = 3$

$\left\{ \frac{1}{2} \rightarrow 1 \right.$

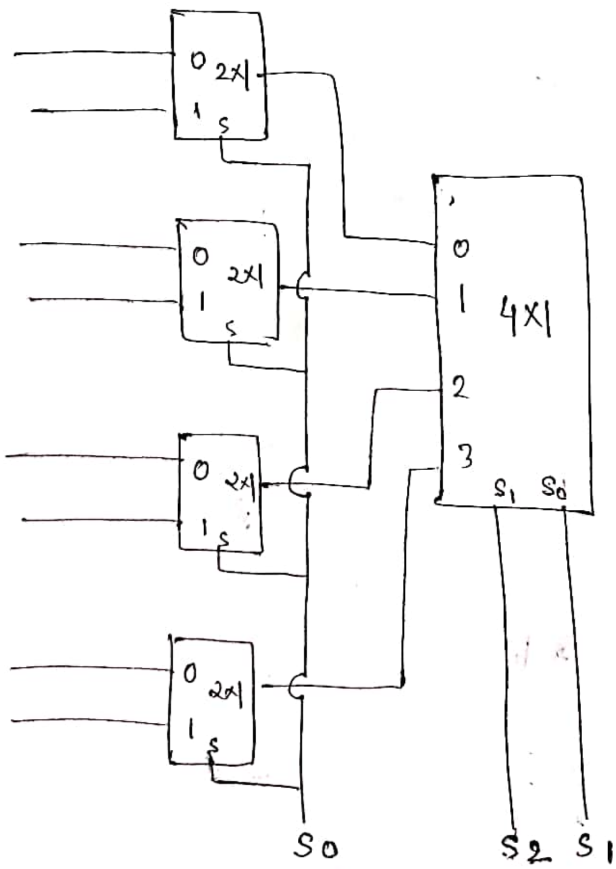
OR



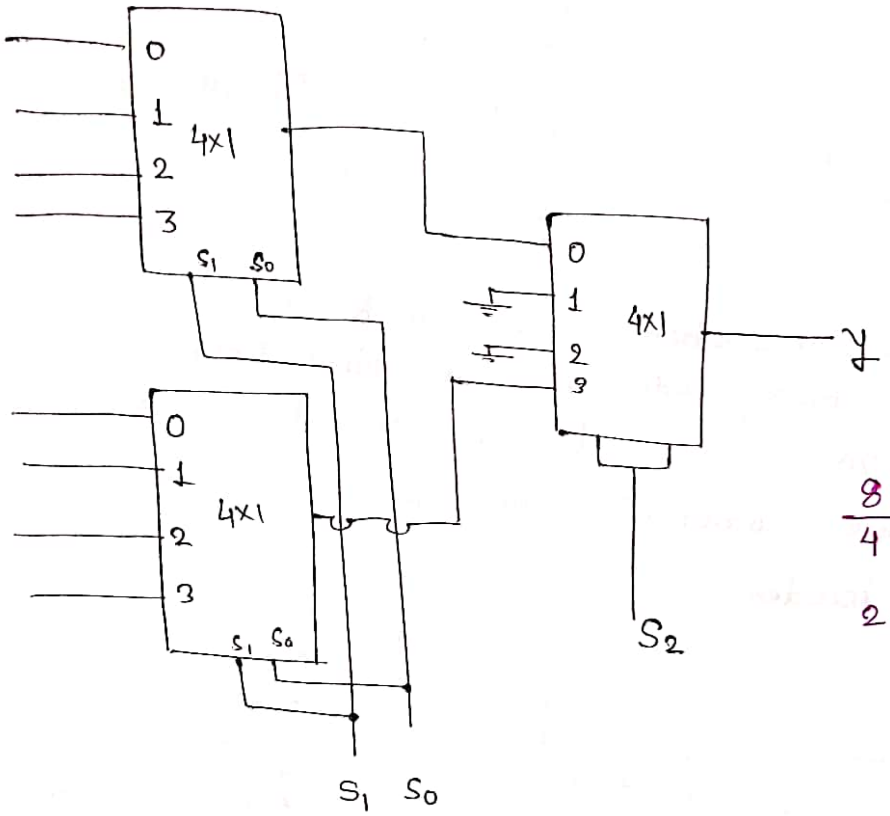
* 8X1 MUX using 4x1 & 2x1



OR



* 8x1 using 4x1

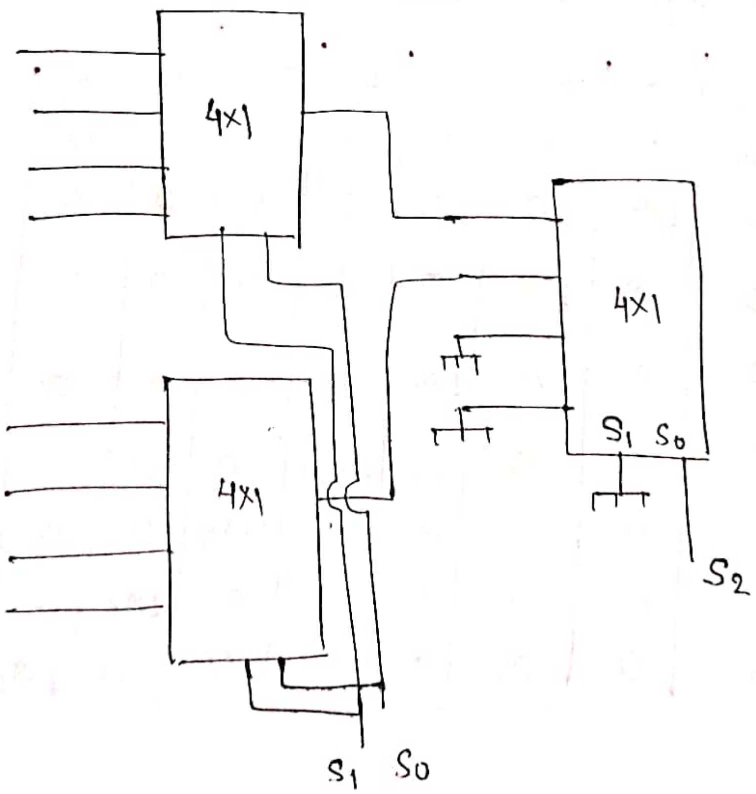


$\frac{8}{4}$ $\frac{2}{4}$

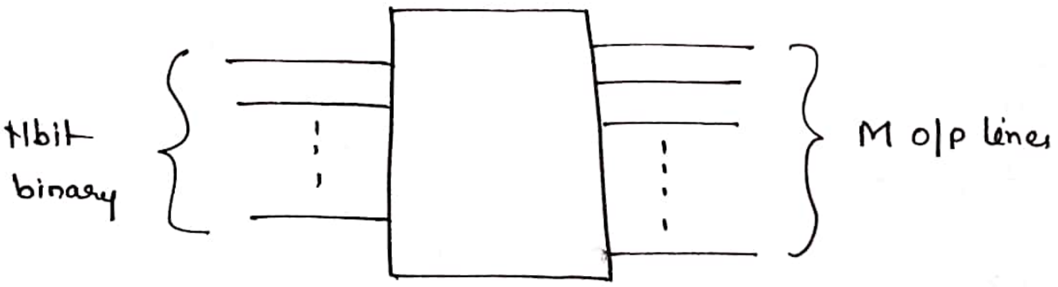
2 1 = 3

$\left\{ \frac{1}{2} \rightarrow 1 \right.$

OR

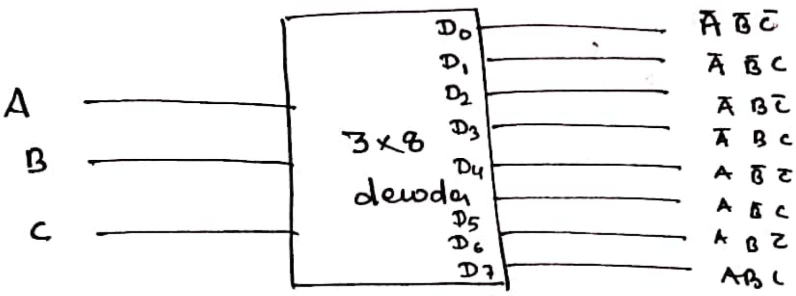


* Decoder



Decoder is a Combinational ckt which Convert n bit Input binary code into M output lines such that one output line is activated for each possible Combination of Input.

eg. 3 x 8 decoder



Truth Table (Active High o/p decoder)

A	B	C	D_0	D_1	D_2	D_3	D_4	D_6	D_6	D_7
0	0	0	1 D_0	0	0	0	0	0	0	0
0	0	1	0	1 D_1	0	0	0	0	0	0
0	1	0	0	0	1 D_2	0	0	0	0	0
0	1	1	0	0	0	1 D_3	0	0	0	0
1	0	0	0	0	0	0	1 D_4	0	0	0
1	0	1	0	0	0	0	0	1 D_5	0	0
1	1	0	0	0	0	0	0	0	1 D_6	0
1	1	1	0	0	0	0	0	0	0	1 D_7

$D_0 = \bar{A}\bar{B}\bar{C}$

$D_1 = \bar{A}\bar{B}C$

$D_2 = \bar{A}B\bar{C}$

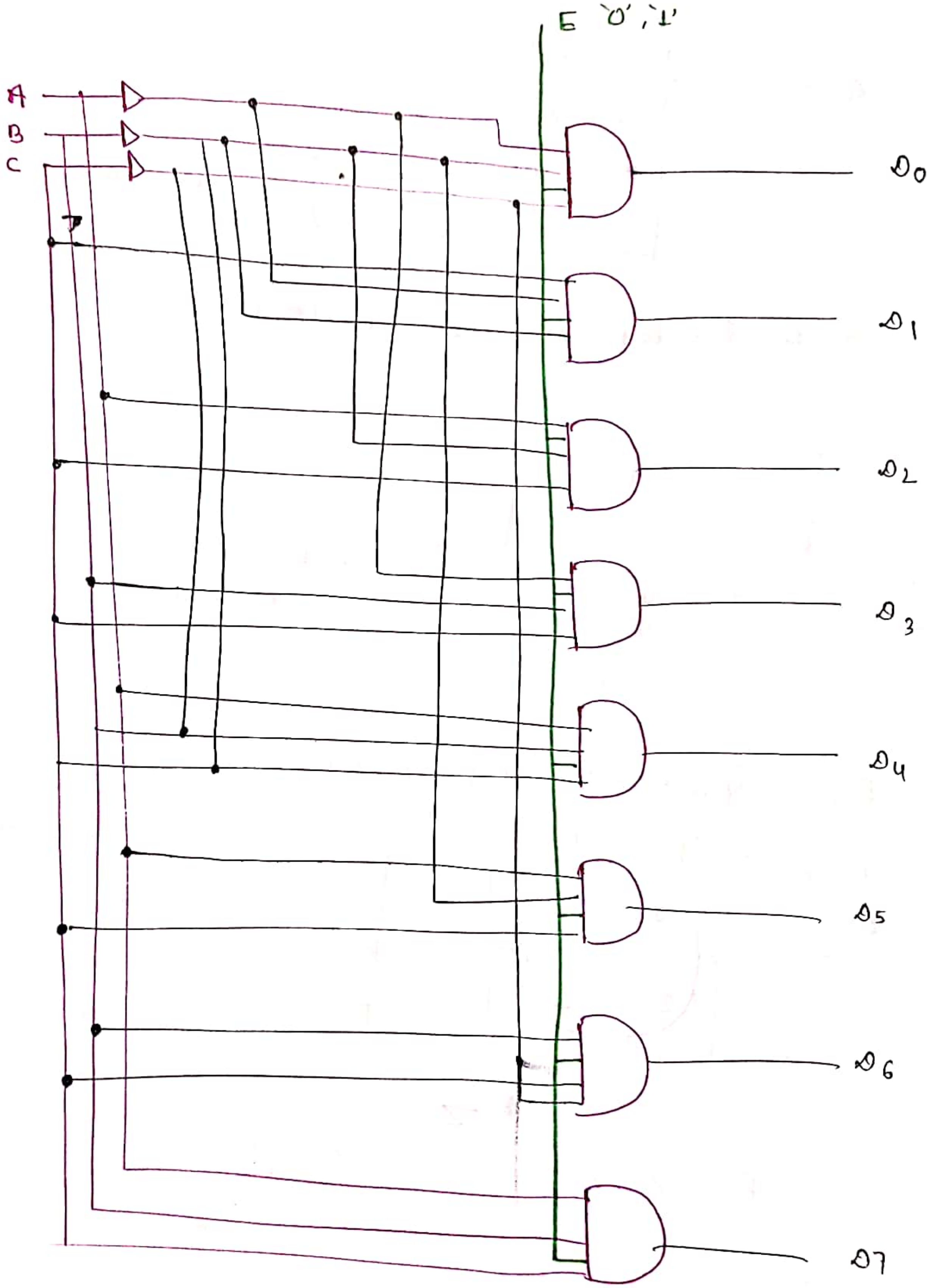
$D_3 = \bar{A}BC$

$D_4 = A\bar{B}\bar{C}$

$D_5 = A\bar{B}C$

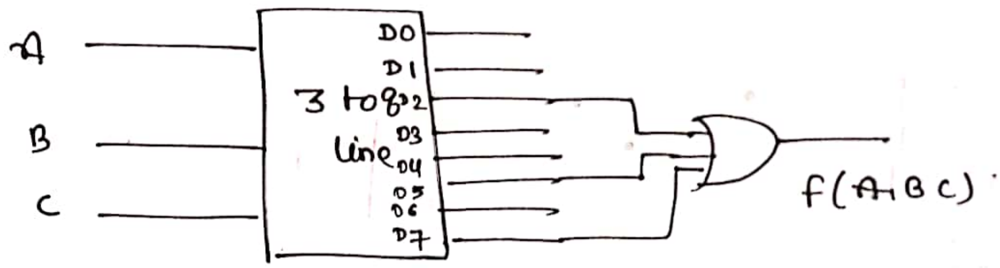
$D_6 = AB\bar{C}$

$D_7 = ABC$



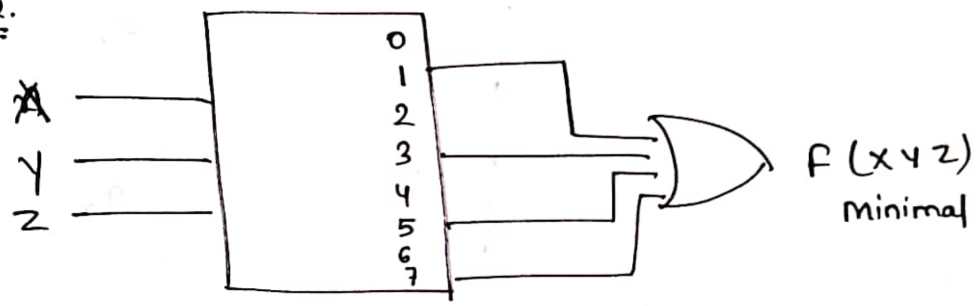
Que. Design $F(A,B,C) = \sum m(2,5,7)$ using 3×8 decoder

Sol $f = \bar{A} B \bar{C} \quad A \bar{B} C \quad A B C$



Note: 3×8 decoder is also a Universal ckt.

Que.



$$f = \bar{x}\bar{y}z + \bar{x}yz + x\bar{y}z + xyz$$

$x \backslash yz$	$\bar{y}\bar{z}$	$\bar{y}z$	yz	$y\bar{z}$
$\bar{x}$		1	1	
x		1	1	

$\rightarrow z$

$F = z$ //

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Que. $f(xyz) = xz + y\bar{z}$ using 3×8 decoder

$= x\bar{y}z + xyz + \bar{x}y\bar{z} + xy\bar{z}$

$= \sum m(2,5,6,7)$

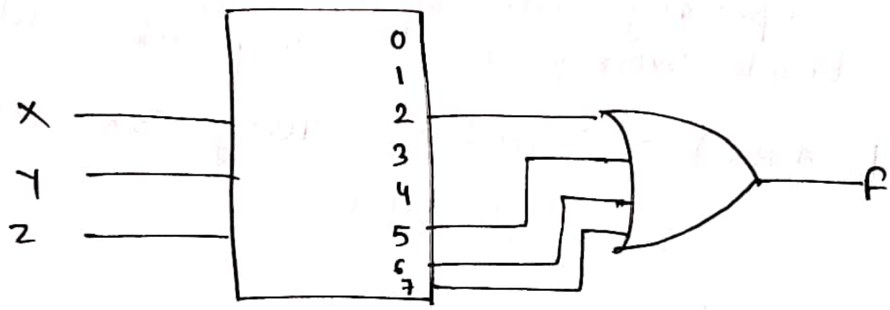
$x \backslash yz$	$\bar{y}\bar{z}$	$\bar{y}z$	yz	$y\bar{z}$
$\bar{x}$				1
x		1	1	1

$y\bar{z}$

yx

xz

$f = xz + xy + y\bar{z}$



Note :- Enable is an extra input of decoder to switch on / off the decoder.

* Decoder with active Low output :

Input			Output							
A	B	C	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7
0	0	0	0	1	1	1	1	1	1	1
0	0	1	1	0	1	1	1	1	1	1
0	1	0	1	1	0	1	1	1	1	1
0	1	1	1	1	1	0	1	1	1	1
1	0	0	1	1	1	1	0	1	1	1
1	0	1	1	1	1	1	1	0	1	1
1	1	0	1	1	1	1	1	1	0	1
1	1	1	1	1	1	1	1	1	1	0

$D_0 = \overline{\bar{A}\bar{B}\bar{C}}$

$D_1 = \overline{\bar{A}\bar{B}C}$

$D_2 = \overline{\bar{A}B\bar{C}}$

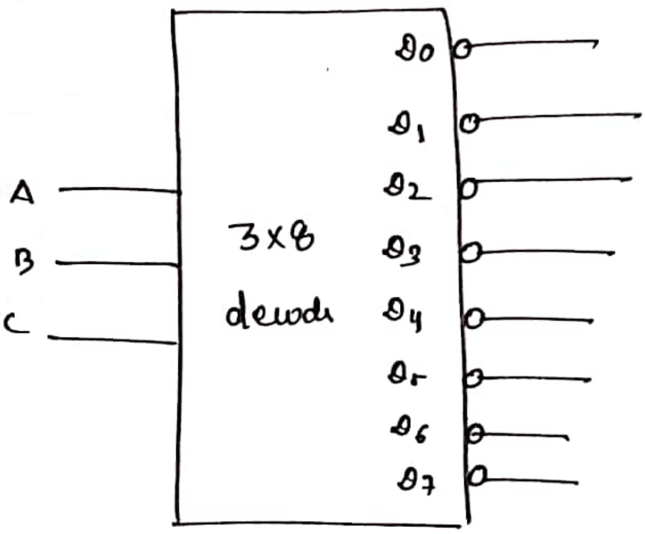
$D_3 = \overline{\bar{A}BC}$

$D_4 = \overline{A\bar{B}\bar{C}}$

$D_5 = \overline{A\bar{B}C}$

$D_6 = \overline{AB\bar{C}}$

$D_7 = \overline{ABC}$



note Circuit of active low o/p decoder is obtain by replacing AND by NAND gate.

* Enable work in a similar way like active high

Que. $F(A,B,C) = \sum m(1,5)$ using 3×8 active low o/p decoder

Sol

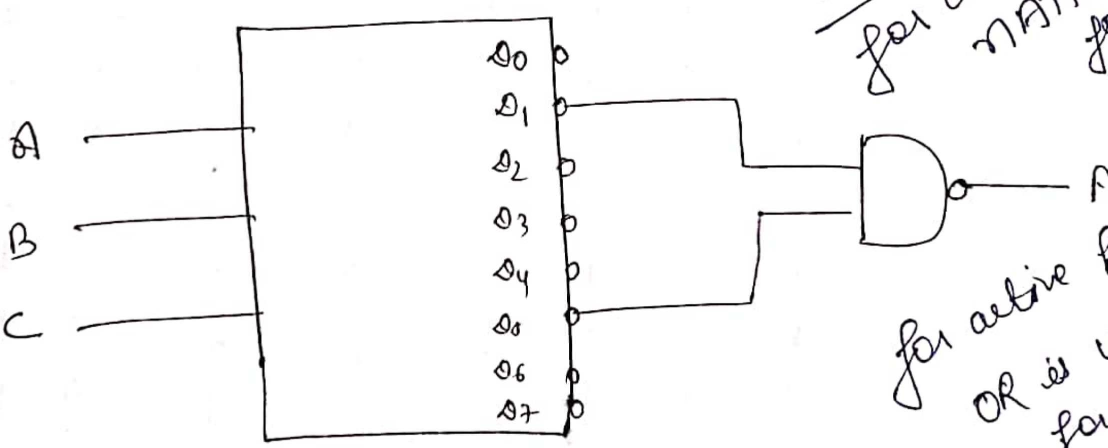
$\sum m(1,5)$

$\sum m(001 \quad 101)$

$\overline{\overline{\bar{A}\bar{B}C + A\bar{B}C}}$

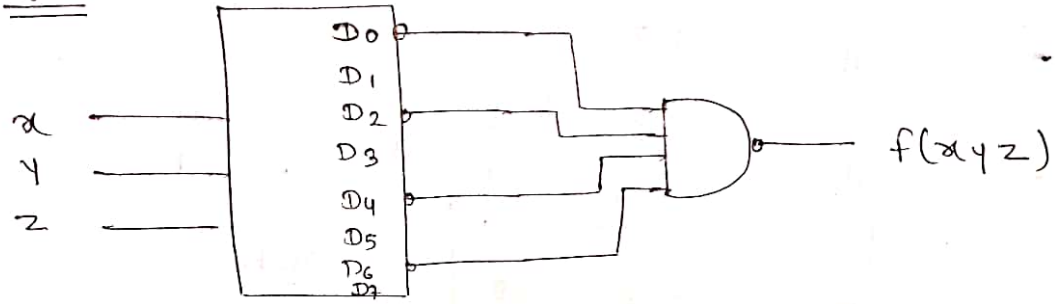
$\Rightarrow \overline{\overline{\bar{A}\bar{B}C} \cdot \overline{A\bar{B}C}}$

note for active low o/p NAND is used for Sum



for active high OR is used for Sum

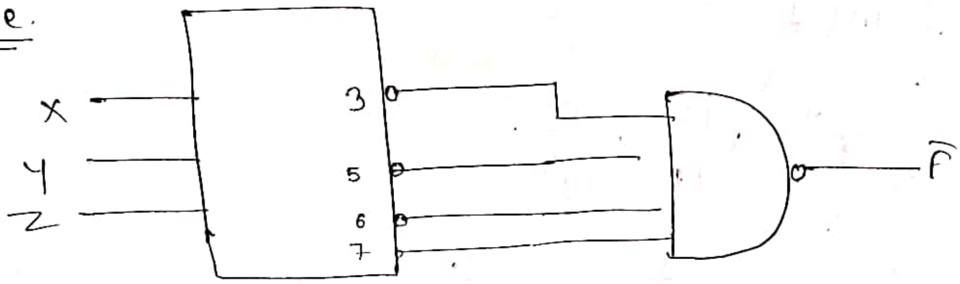
Que.



Sol ~~$\bar{A}\bar{B}\bar{C}$~~

$$f(x,y,z) = \sum m(0\ 2\ 4\ 6)$$

Que.

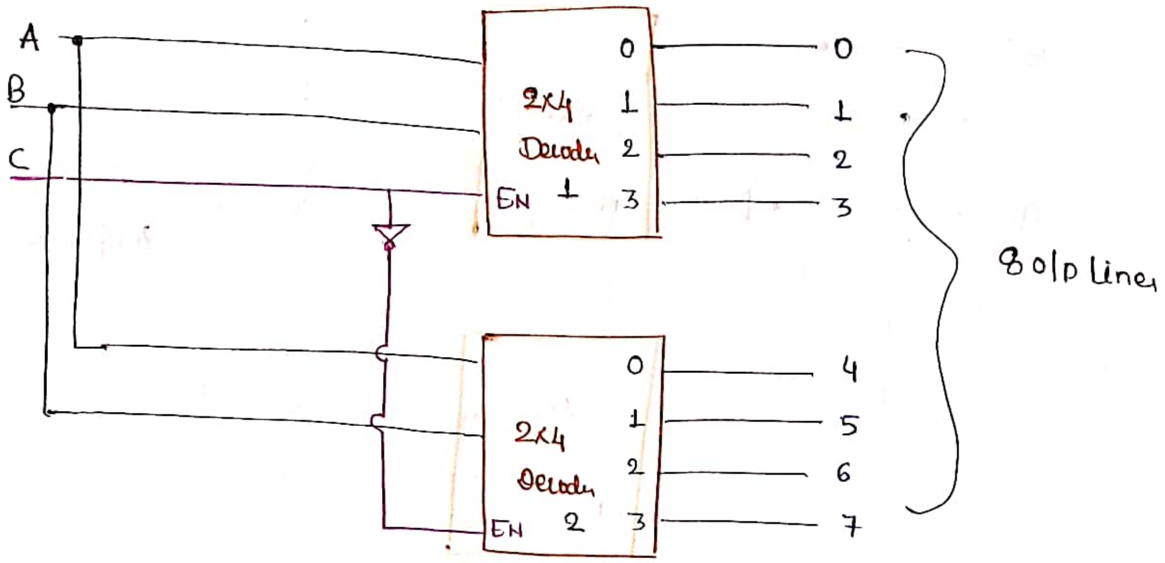


$$f = \sum m(3,5,6,7)$$

x \ yz		$\bar{y}z$	yz	$y\bar{z}$
$\bar{x}$			1	
x		1	1	1

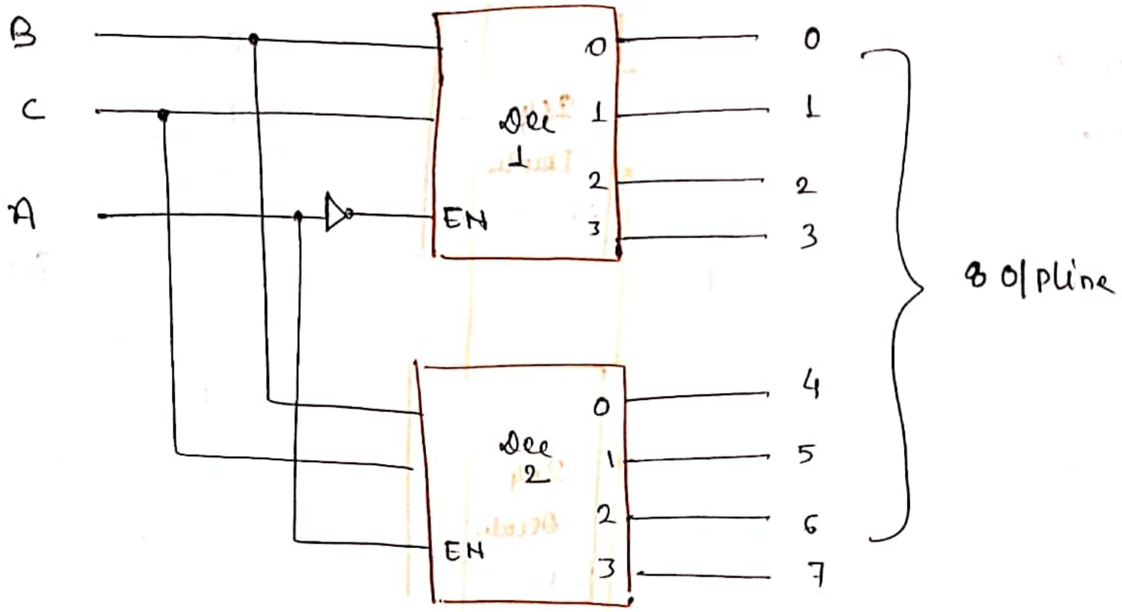
$$f = yz + xz + xy$$ //

Que. Implement 3x8 decoder using Minimum no. of 2x4 decoder. Enable input are avaliable in all 2x4 decoders. Use an extra Inverter if required.



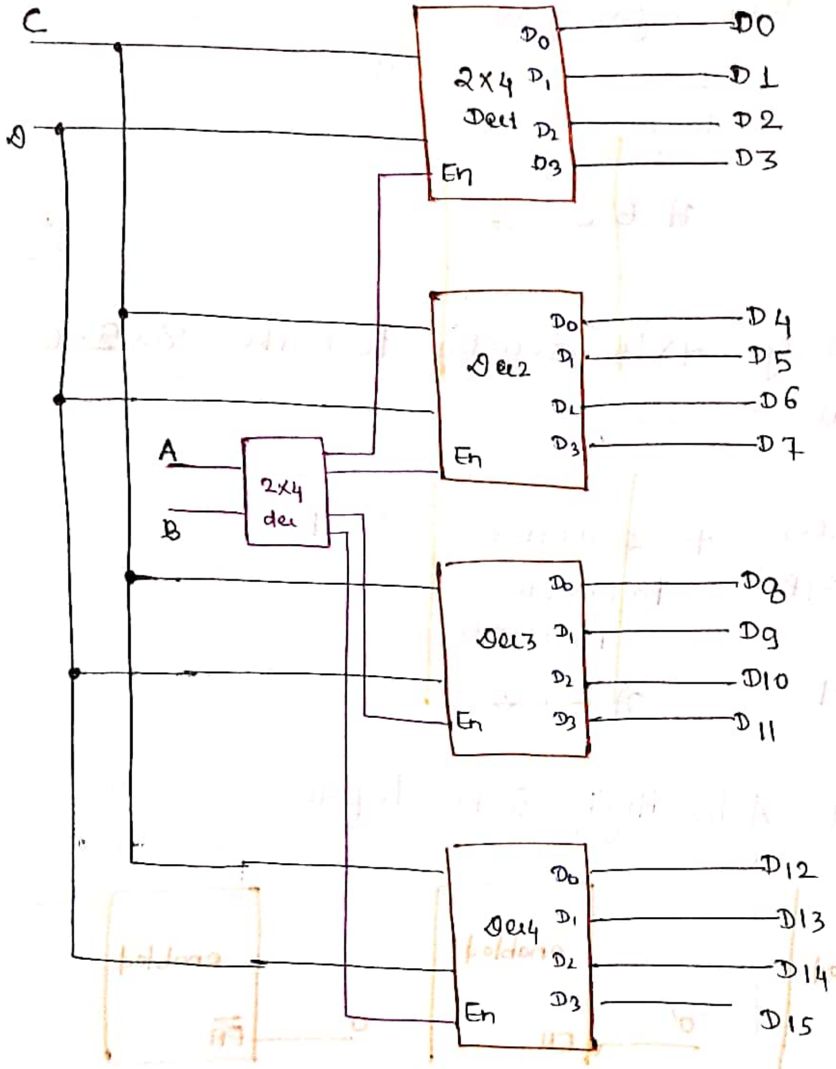
A	B	C	Enable Decoder	Activated line
0	0	0	Dec. 2	D_4
0	0	1	Dec 1	D_0
0	1	0	Dec 2	D_5
0	1	1	Dec 1	D_1
1	0	0	Dec 2	D_6
1	0	1	Dec 1	D_2
1	1	0	Dec 2	D_7
1	1	1	Dec 1	D_3

but this is not In Sequence



A	B	C	Enable Dec	Activated line
0	0	0	Dec1	D_0
0	0	1	Dec1	D_1
0	1	0	Dec1	D_2
0	1	1	Dec1	D_3
1	0	0	Dec2	D_4
1	0	1	Dec2	D_5
1	1	0	Dec2	D_6
1	1	1	Dec2	D_7

Que Design 4x16 Using 2x4



note
Every o/p is Multiplied by Enable in decoder

A	B	C	D	enabled decoder	Activated line
0	0	0	0	Dec 1	D0
0	0	0	1		D1
0	0	1	0		D2
0	0	1	1		D3
0	1	0	0	Dec 2	D4
0	1	0	1		D5
0	1	1	0		D6
0	1	1	1		D7
1	0	0	0	Dec3	D8
1	0	0	1		D9
1	0	1	0		D10
1	0	1	1		D11
1	1	0	0	Dec4	D12
1	1	0	1		D13
1	1	1	0		D14
1	1	1	1		D15

Que. How Many 3 line to 8 line decoders with Enable i/p are needed to Construct 6 line to 64 line decoder without any other logic gates

Ans: 8 decoder for 64 o/p + 1 decoder to Enable 8 decoder = 9

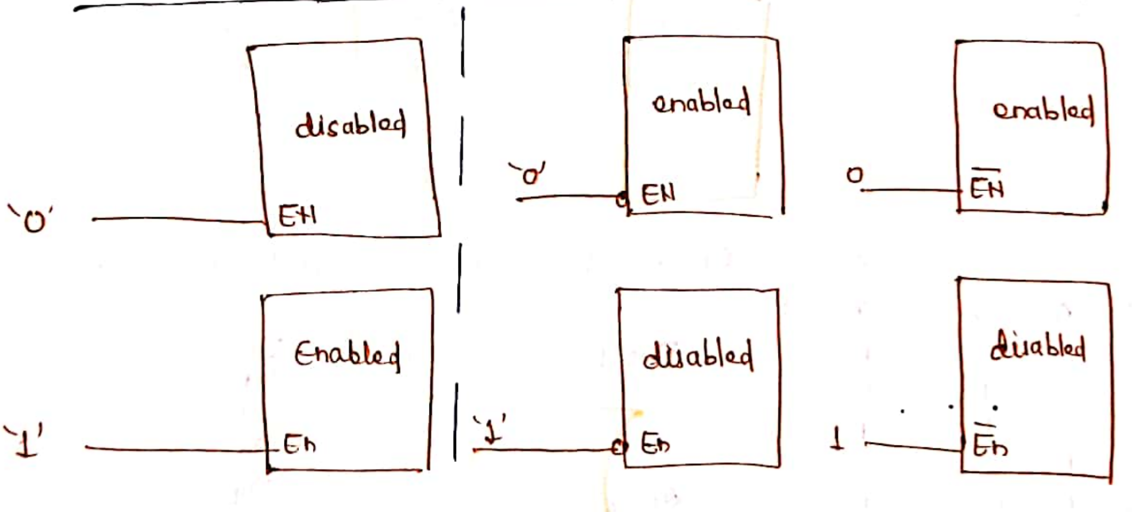
DEF   A B C

Que. The Min. no. of 4×16 decoder to make 8×256 decoder is ______

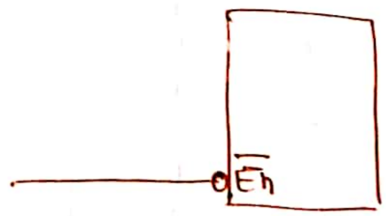
Sol 16 decoder for 256 o/p + 1 decoder to Enable 16 decoder = 17

E F G H   A B C D

* **Active low and Active High Enable Inputs**



Active High Enable   Active low Enable



* ENCODER

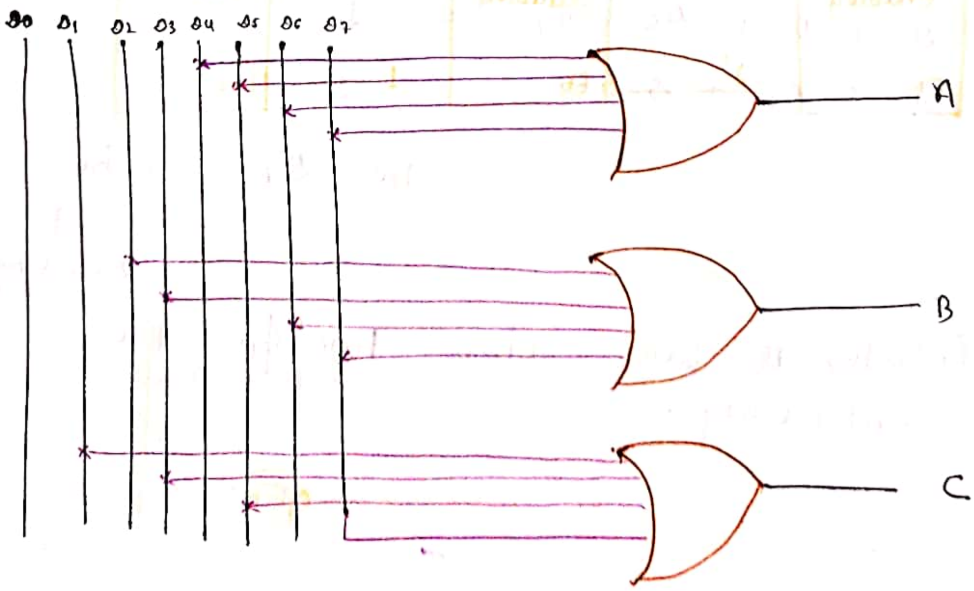
→ Encoder is a Combinational circuit which perform the reverse operation of a Decoder

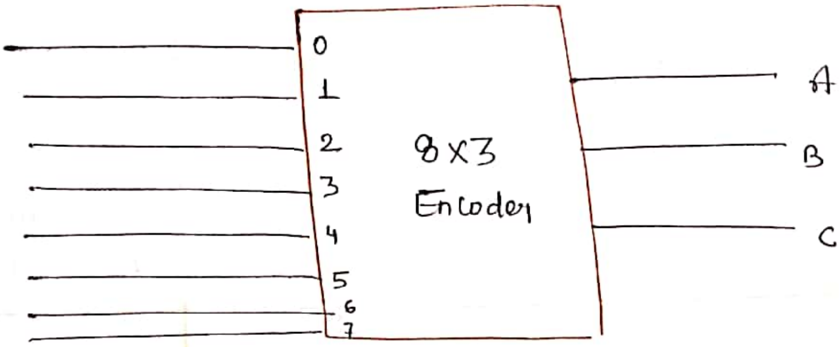
Inputs								Outputs		
D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	A	B	C
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

$A = D_4 + D_5 + D_6 + D_7$

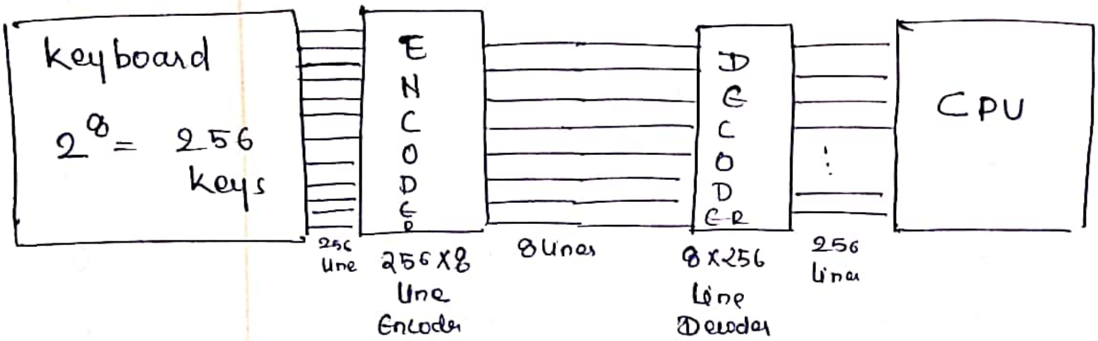
$B = D_2 + D_3 + D_6 + D_7$.

$C = D_1 + D_3 + D_5 + D_7$

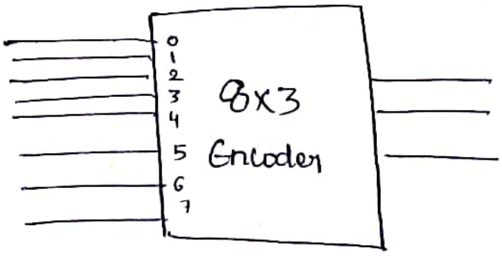




example



If both 3 & 4 active at a time :



$A = D_4 + D_5 + D_6 + D_7 = 1$

$B = D_2 + D_3 + D_6 + D_7 = 1$

$C = D_1 + D_3 + D_5 + D_7 = 1$

i.e D_7 will be recognized after Decoding

Priority Encoder is used when two i/p active Simultaneously.

* Priority Encoder

D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	A	B	C
1	0	0	0	0	0	0	0	0	0	0
X	1	0	0	0	0	0	0	0	0	1
X	X	1	0	0	0	0	0	0	1	0
X	X	X	1	0	0	0	0	0	1	1
X	X	X	X	1	0	0	0	1	0	0
X	X	X	X	X	1	0	0	1	0	1
X	X	X	X	X	X	1	0	1	1	0
X	X	X	X	X	X	X	1	1	1	1

8 X 3 priority Encoder

$$A = D_4 \overline{D_5}\,\overline{D_6}\,\overline{D_7} + D_5 \overline{D_6}\,\overline{D_7} + D_6 \overline{D_7} + D_7$$

$$B = D_2 \overline{D_3}\,\overline{D_4}\,\overline{D_5}\,\overline{D_6}\,\overline{D_7} + D_3 \overline{D_4}\,\overline{D_5}\,\overline{D_6}\,\overline{D_7} + D_6 \overline{D_7} + D_7$$

$$C = D_1 \overline{D_2}\,\overline{D_3}\,\overline{D_4}\,\overline{D_5}\,\overline{D_6}\,\overline{D_7} + D_3 \overline{D_4}\,\overline{D_5}\,\overline{D_6}\,\overline{D_7} + D_5 \overline{D_6}\,\overline{D_7} + D_7$$

* 4 X 2 priority Encoder

D_0	D_1	D_2	D_3	A	B
1	0	0	0	0	0
X	1	0	0	0	1
X	X	1	0	1	0
X	X	X	1	1	1

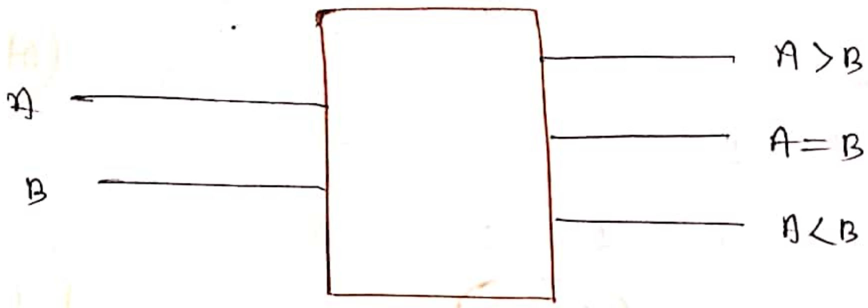
$$A = D_4 \overline{D_5}\,\overline{D_6}\,\overline{D_7} + D_5 \overline{D_6}\,\overline{D_7} + D_6 \overline{D_7} + D_7$$

$$= (\overline{D_6}\,\overline{D_7} + D_4 \overline{D_5})(\overline{D_7} + D_6) + D_5 \overline{D_6}\,\overline{D_7} + D_4 \overline{D_5}\,\overline{D_6}\,\overline{D_7}$$

$$= D_4 + D_5 + D_6 + D_7$$

* Comparator

→ It is used to Compare two Input numbers

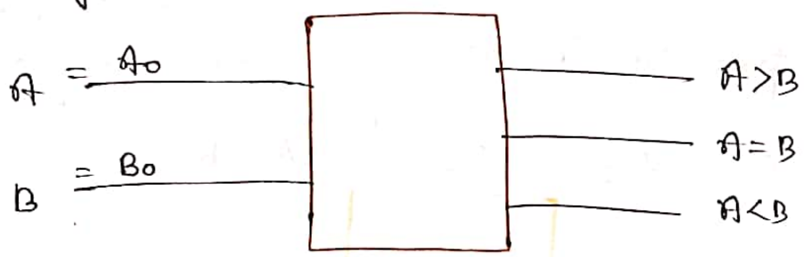


Comparator

→ Comparator has 3 o/p. namely A>B , A=B , A>B

• 1 bit Comparator

→ One bit Comparator should be able to Compare two Inputs of 1bit each



1 bit Comp.

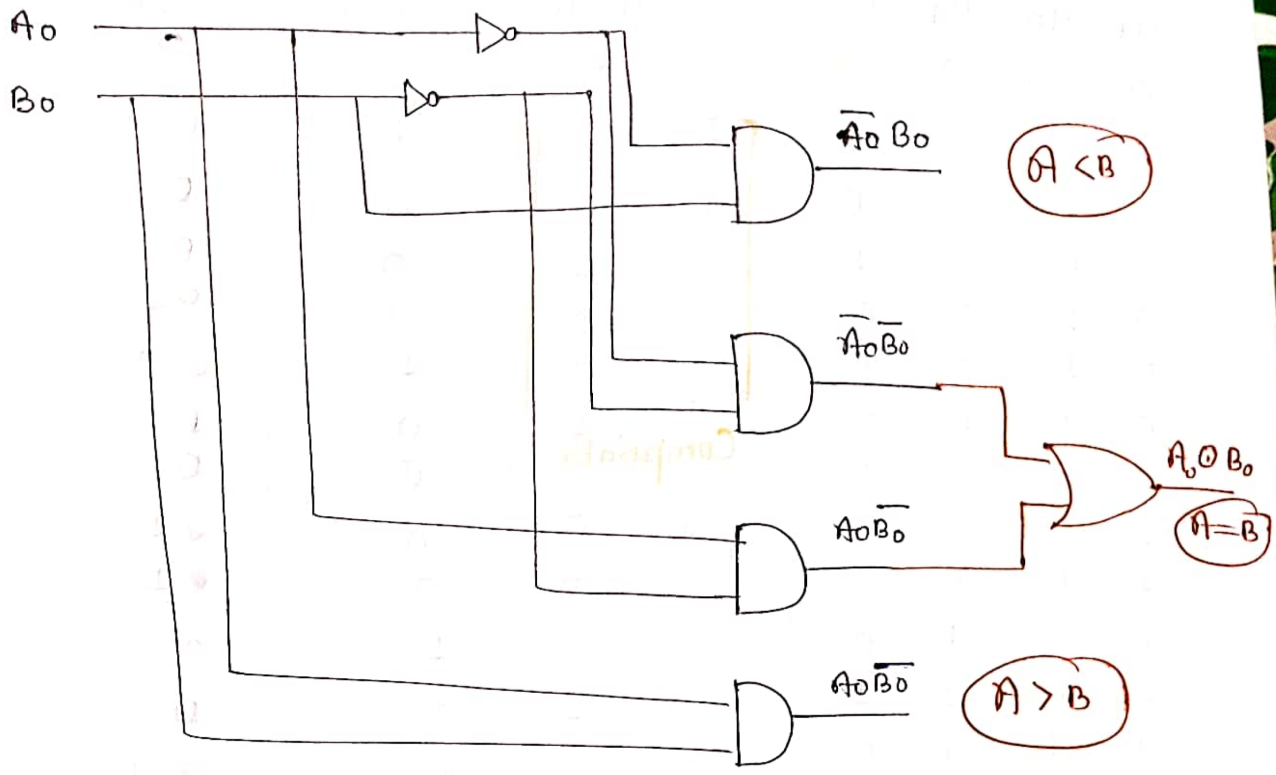
Input		Output		
A_0	B_0	$A<B$	$A=B$	$A>B$
0	0	0	1	0
0	1	1	0	0
1	0	0	0	1
1	1	0	1	0

$A<B = \overline{A_0} B_0$

$A=B = \overline{A_0}\,\overline{B_0} + A_0 B_0 = A_0 \odot B_0$

$A>B = A_0 \overline{B_0}$

note
Must Remember



* 2 bit Comparator

→ Two bit Comparator should be able to compare two i/p numbers of two bits each

2 bit Comparator

A_1	A_0	B_1	B_0	$A<B$	$A=B$	$A>B$
0	0	0	0	0	1	0
0	0	0	1	1	0	0
0	0	1	0	1	0	0
0	0	1	1	1	0	0
0	1	0	0	0	0	1
0	1	0	1	0	1	0
0	1	1	0	1	0	0
0	1	1	1	1	0	0
1	0	0	0	0	0	1
1	0	0	1	0	0	1
1	0	1	0	0	1	0
1	0	1	1	1	0	0
1	1	0	0	0	0	1
1	1	0	1	0	0	1
1	1	1	0	0	0	1
1	1	1	1	0	1	0

for $A<B$:

$A_1A_0 \backslash B_1B_0$		$\bar{B_1}B_0$	B_1B_0	$B_1\bar{B_0}$
$\bar{A_1}\bar{A_0}$		1	1	1
$\bar{A_1}A_0$			1	1
$A_1\bar{A_0}$			1	

$$A<B = \bar{A_1}B_1 + \bar{A_1}\bar{A_0}B_0 + \bar{A_0}B_1B_0$$

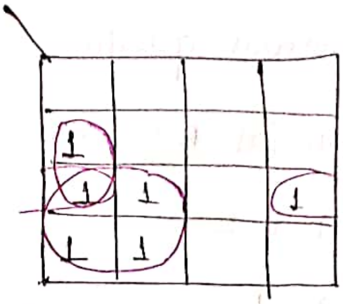
for $A=B$

$A A_0 \backslash B_1B_0$				
	1			
		1		
			1	
				1

$$A=B = \bar{A_1}\bar{A_0}\bar{B_1}\bar{B_0} + \bar{A_1}A_0\bar{B_1}B_0 + A_1A_0B_1B_0 + A_1\bar{A_0}B_1\bar{B_0}$$

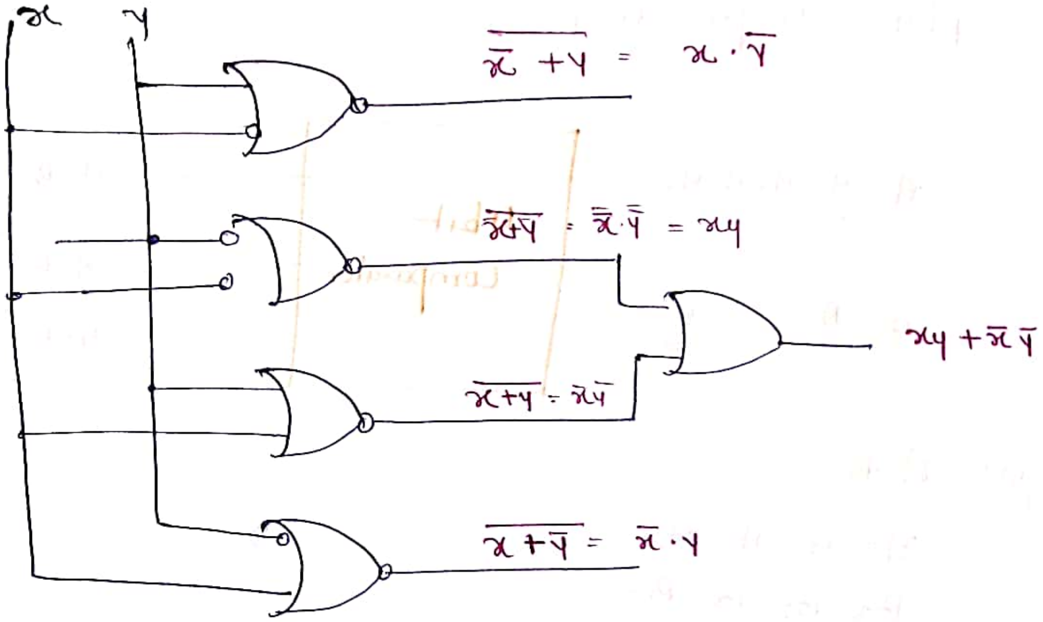
$$= (A_1 \odot B_1)(A_0 \odot B_0)$$

for A>B



$A>B = A_1\bar{B_1} + A_0\bar{B_1}\bar{B_0} + A_1A_0\bar{B_0}$

Que. Identify the given Circuit



Sol $x\cdot\bar{y} = y < x$

$xy+\bar{x}\bar{y} = y = x$

$y\bar{x} = y > x$

Que. The o/p 'Y' of a two bit Comparator is '1' whenever the two bit i/p A is greater than two bit i/p B. The no. of combⁿ for which the o/p is logic 1 is ____

Sol Six //.

Q. for n bit Comparator in previous question

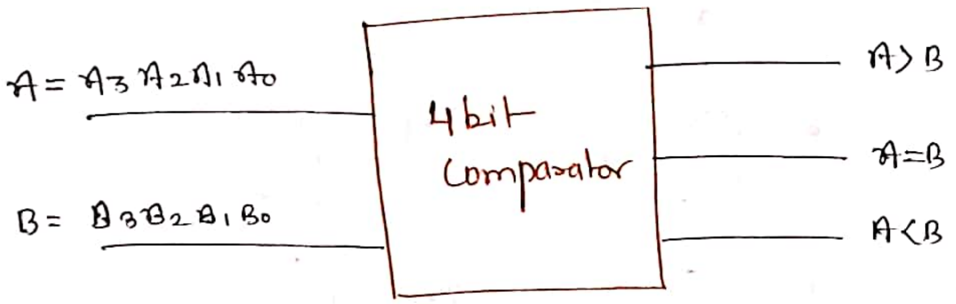
Sol Total = no.(A>B) + no.(A=B) + no.(A<B)

$2^{2n} = 2\ \text{no.}(A>B) + 2^{n}$

note

$\text{no.}(A>B) = 2^{2n-1} - 2^{n-1}$ //.

* 4 bit Comparator



for: $A>B$

$A_3\ A_2\ A_1\ A_0$

$B_3\ B_2\ B_1\ B_0$

$A_3 > B_3$ OR

$A_3 = B_3$ & $A_2 > B_2$ OR

$A_3 = B_3$ & $A_2 = B_2$ & $A_1 > B_1$ OR

$A_3 = B_3$ & $A_2 = B_2$ & $A_1 = B_1$ & $A_0 > B_0$

$$\Rightarrow \boxed{A>B = A_3\overline{B_3} + (A_3 \odot B_3)(A_2\overline{B_2}) + (A_3 \odot B_3)(A_2 \odot B_2)(A_1\overline{B_1}) + (A_3 \odot B_3)(A_2 \odot B_2)(A_1 \odot B_1)(A_0\overline{B_0})}$$

Scanned by CamScanner

for $A < B$

$$\boxed{A<B = \bar{A}_3 B_3 + (A_3 \odot B_3)(\bar{A}_2 B_2) + (A_3 \odot B_3)(A_2 \odot B_2)\bar{A}_1 B_1 + (A_3 \odot B_3)(A_2 \odot B_2)(A_1 \odot B_1)\bar{A}_0 B_0}$$

$$\boxed{\text{for } A=B \quad = (A_3 \odot B_3) * (A_2 \odot B_2)(A_1 \odot B_1)(A_0 \odot B_0)}$$

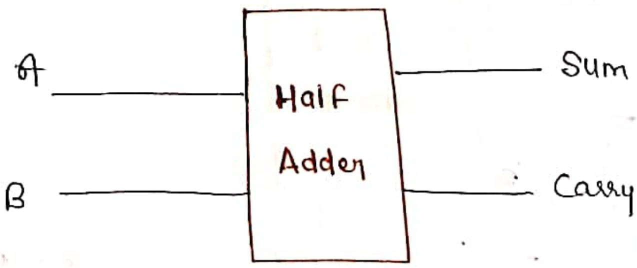
$\cancel{\bar{A}_3}$ $\cancel{\bar{A}_2}$

* Half Adder

→ Half adder is a Combinational ckt which is used for the addition of two binary bits.

0	0	1	1
+ 0	+ 1	+ 0	+ 1
0	1	1	1 0
			carry 1

→ Half adder has two o/p namely Sum and Carry.



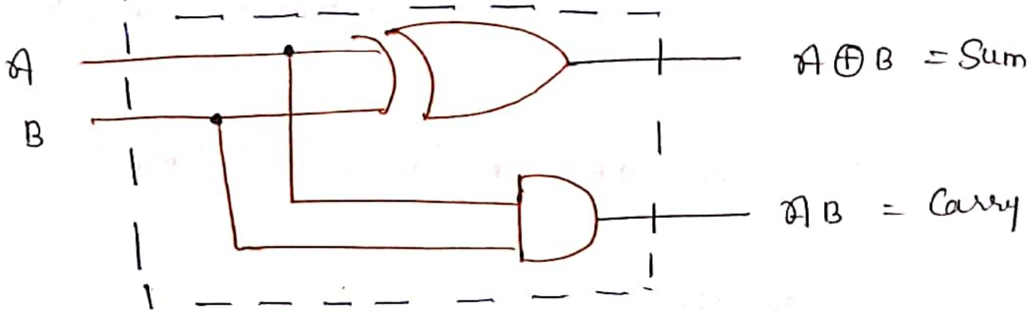
Table

Inputs		Outputs	
A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Logic eq^n

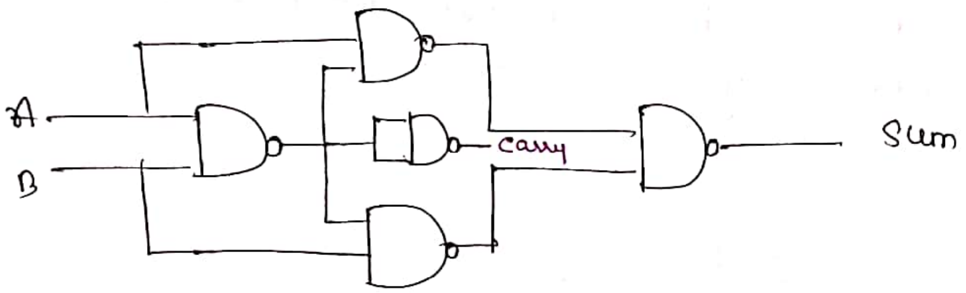
Sum = $A \oplus B$

Carry = AB



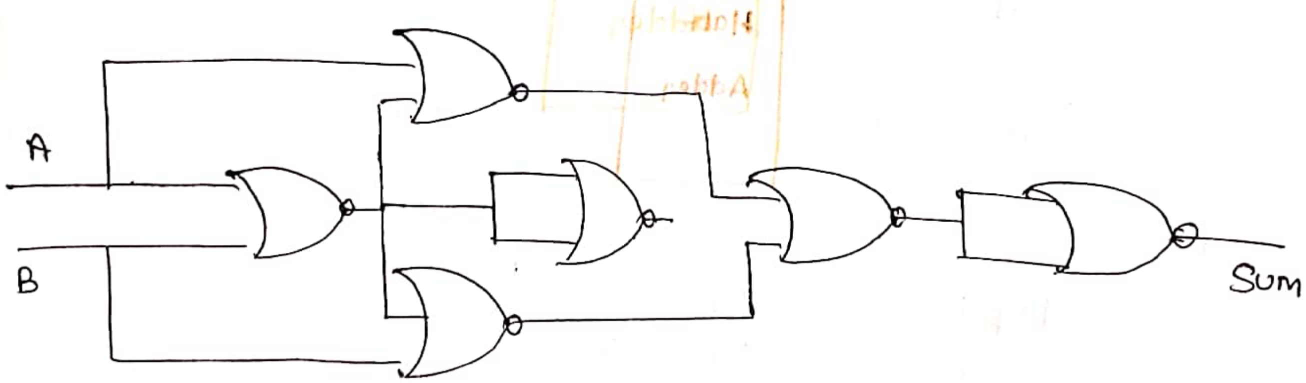
Half adder Using NAND gate

Sum = $\bar{A}B + A\bar{B}$



Five NAND gate required for Half adder.

Half adder using NOR gate

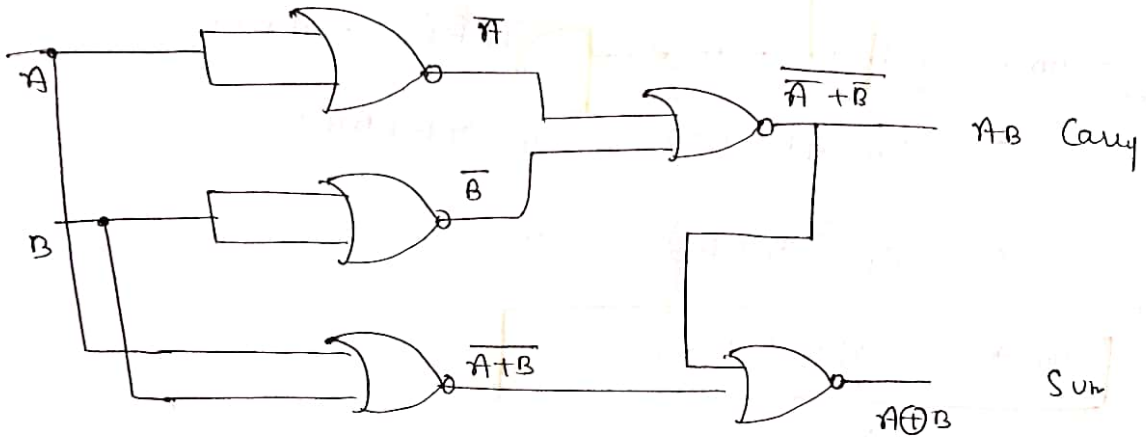


Wrong

$$Sum = \overline{\overline{(A+B)(\bar{A}+\bar{B})}}$$

$$= \overline{\overline{A+B} + \overline{\bar{A}+\bar{B}}}$$

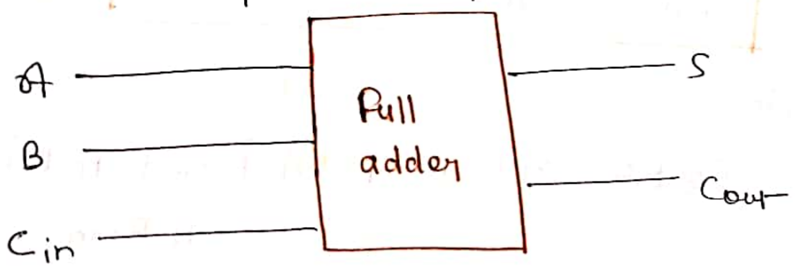
$$=$$



* Full adder

→ It is a Combinational ckt which is used for addition of two Binary bit along with an Input Carry.

→ It has two outputs namely Sum & Carry.



Table

Input			Output	
A	B	Cin	S	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

equation

Sum

A \ BCin	$\bar{B}\bar{C}_{in}$	$\bar{B}C_{in}$	BC_{in}	$B\bar{C}_{in}$
$\bar{A}$	0	1	0	1
A	1	0	1	

$$Sum = \bar{A}\,\bar{B}\,C_{in} + \bar{A}\,B\,\bar{C}_{in} + A\,\bar{B}\,\bar{C}_{in} + A\,B\,C_{in}$$

$$= C_{in}(\bar{A}\bar{B} + AB) + \bar{C}_{in}(\bar{A}B + A\bar{B})$$

$$= C_{in}(\overline{A \oplus B}) + \bar{C}_{in}(A \oplus B)$$

$$\boxed{Sum \Rightarrow A \oplus B \oplus C_{in}}$$

Carry :-

A \ BCin	$\bar{B}\bar{C}_{in}$	$\bar{B}C_{in}$	BC	$B\bar{C}_{in}$
$\bar{A}$			1	
A		1	1	1

$$Carry\ out = B\,C_{in} + A\,C_{in} + A\,B$$

$$\boxed{C_{out} = AB + A\,C_{in} + B\,C_{in}}$$

OR by minterm

$$C_{out} = \bar{A}\,B\,C_{in} + A\,\bar{B}\,C_{in} + A\,B\,\bar{C}_{in} + A\,B\,C_{in}$$

$$= C_{in}(\bar{A}B + A\bar{B}) + AB$$

$$\boxed{C_{out} \Rightarrow C_{in}(A \oplus B) + AB}$$

note. $\Sigma m(1\ 2\ 4\ 7) = A \oplus B \oplus C$

$F(A\,B\,C) =$

Design fulladder using NAND gate only

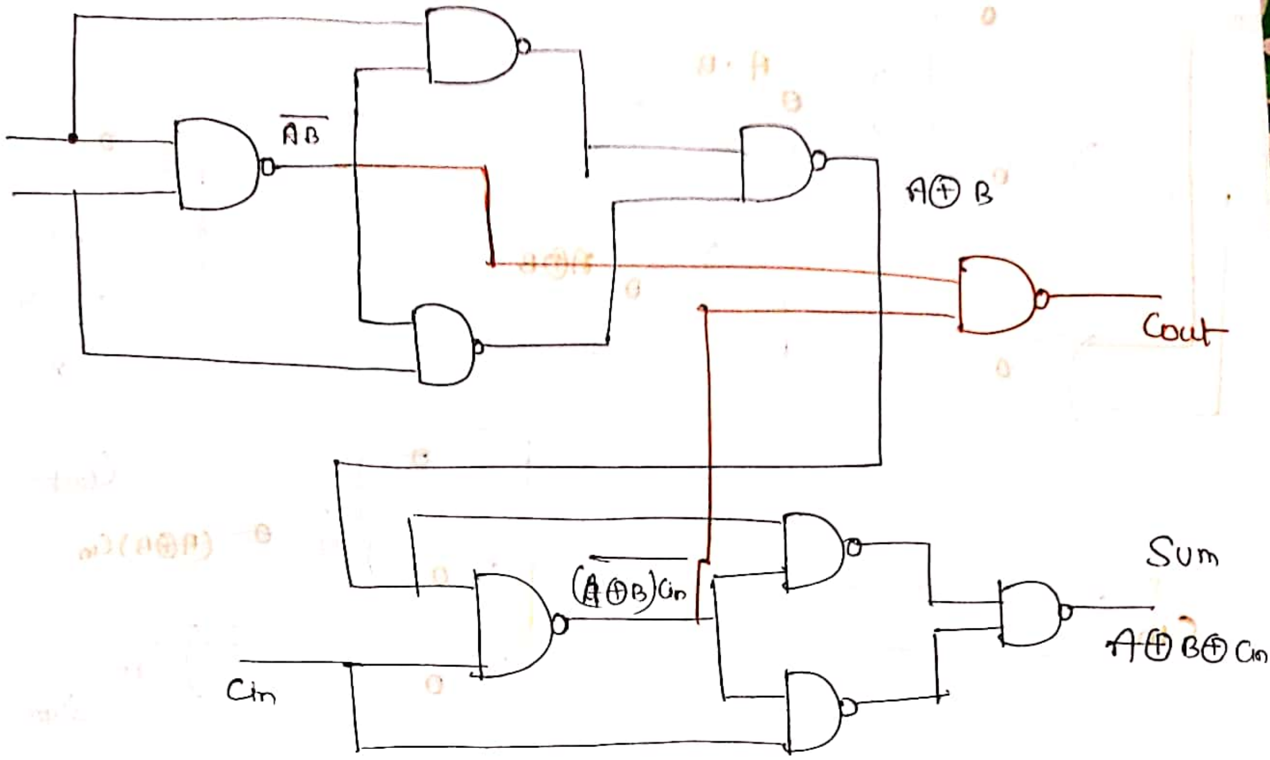


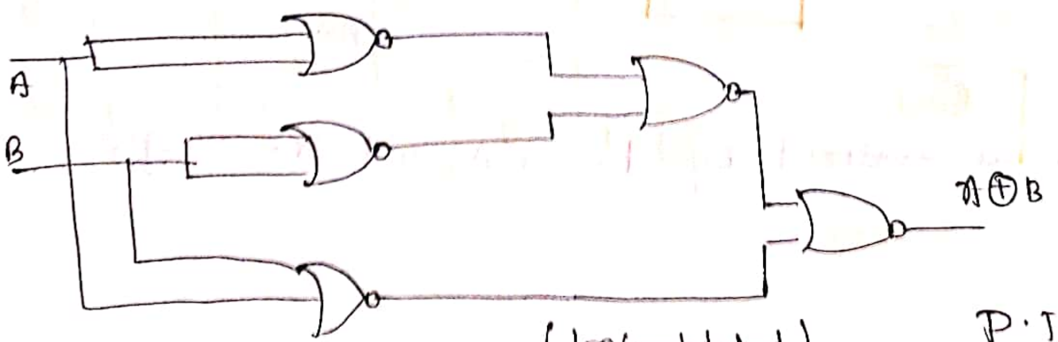
fig. Sum logic of fulladder

now. $C_{out} = AB + (A \oplus B) C_{in}$

Total 9 NAND gate for full adder

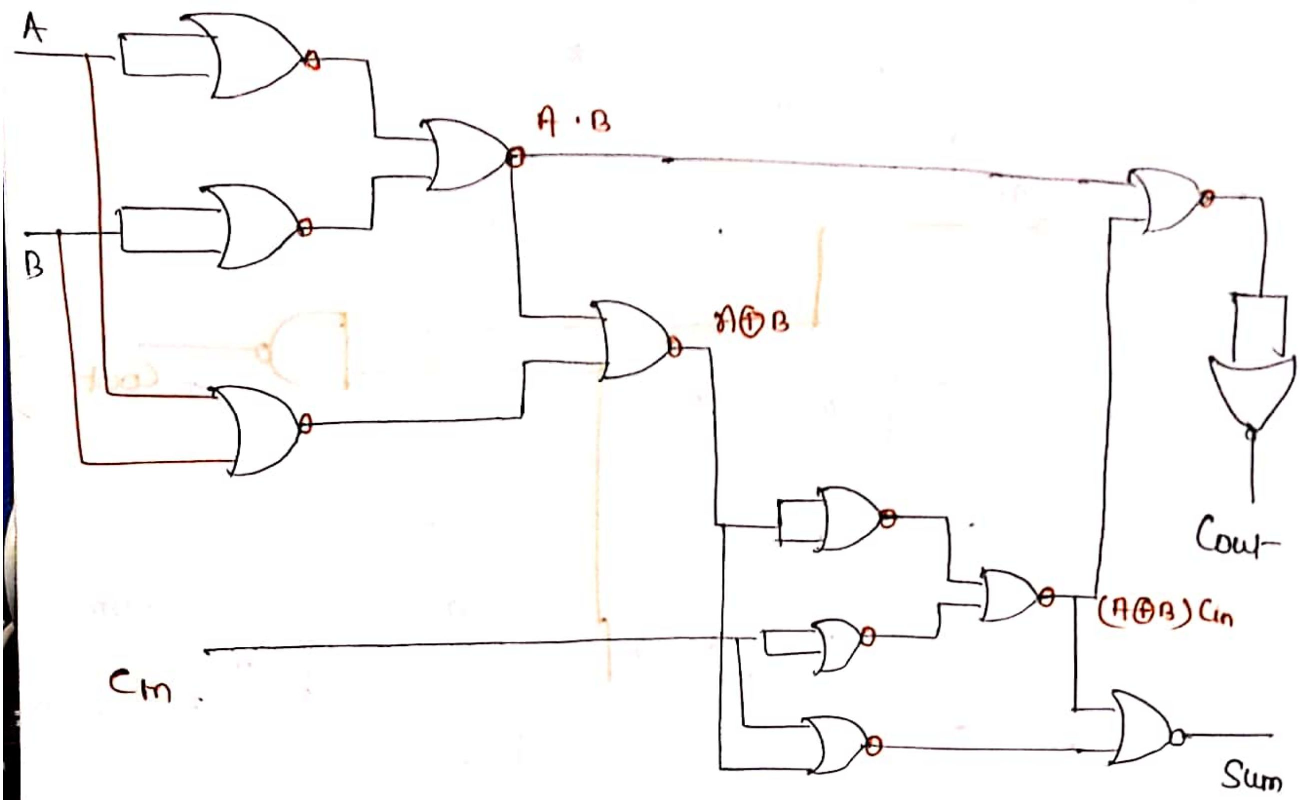
* Design full adder using NOR gate

$$A \oplus B = \bar{A}B + A\bar{B} = \overline{\overline{(A+B)(\bar{A}+\bar{B})}} = \overline{\overline{A+B} + \overline{\bar{A}+\bar{B}}}$$

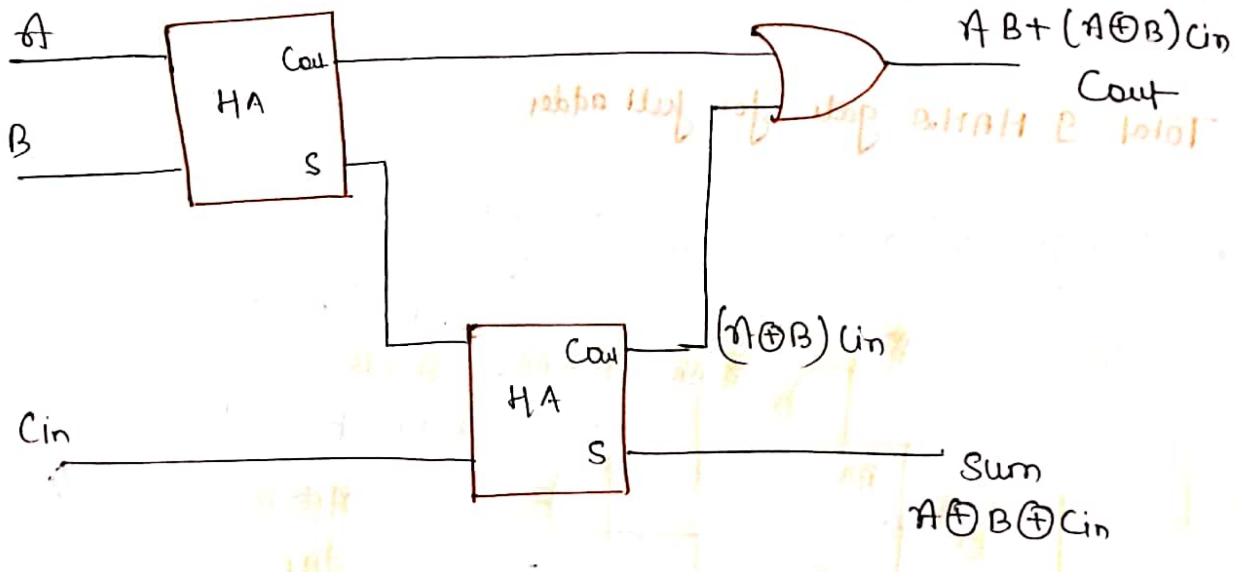


(Incompleted)

P.T.O



* Implement full adder Using Half adder :-

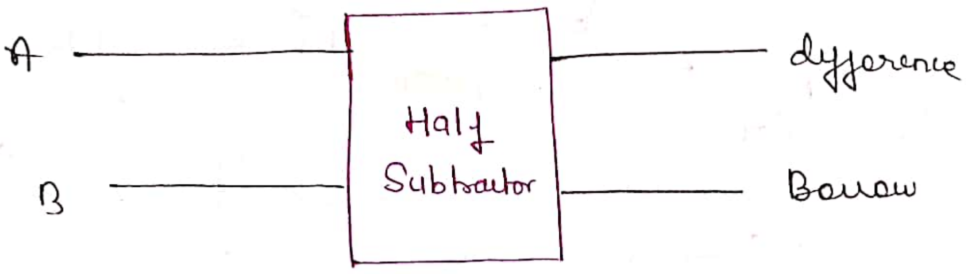


Full adder is realized by two HA and one OR gate

* Half Substractor

→ It is a Combinational ckt used to find difference of two binary bits.

→ It has two o/p namely difference & borrow.

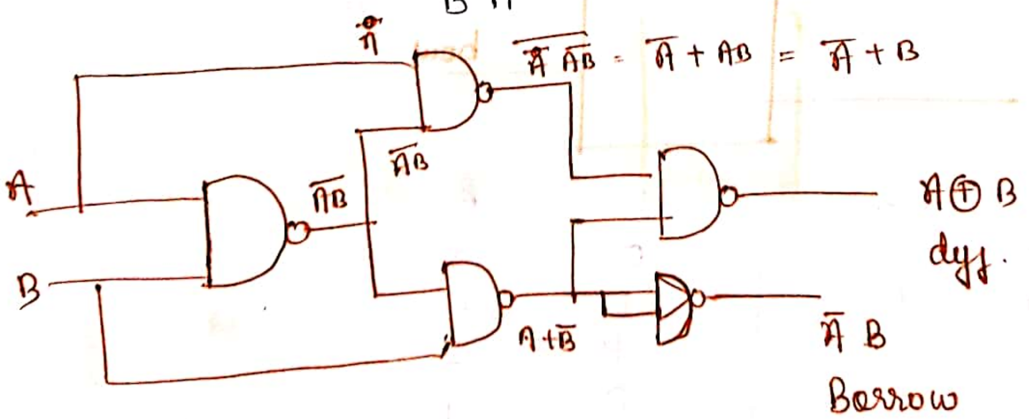


Inputs		Outputs	
A	B	Difference	Borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Difference = $A \oplus B$

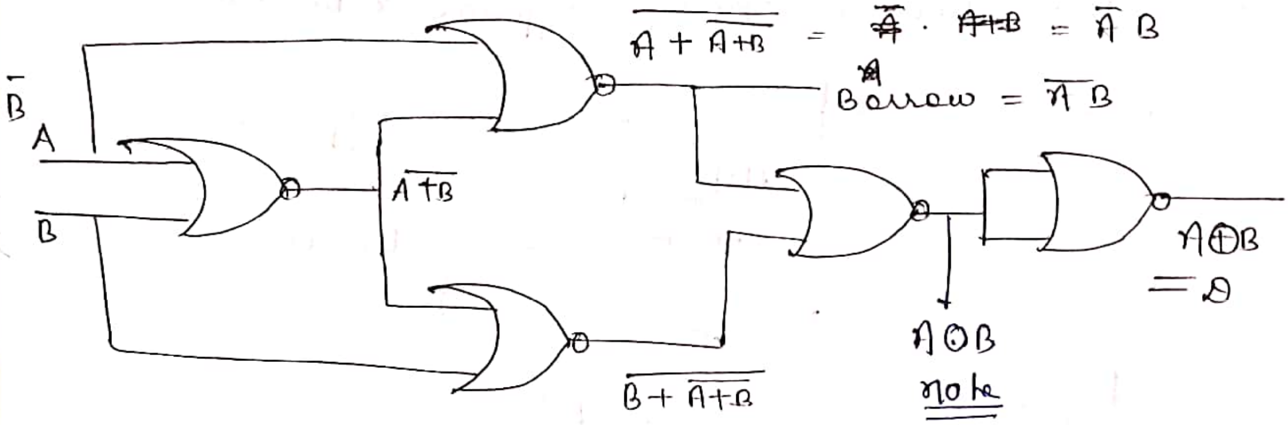
Borrow = $\bar{A} B$ when $A - B$

$\bar{B} A$ when $B - A$



· five NAND gate for Half Subtractor

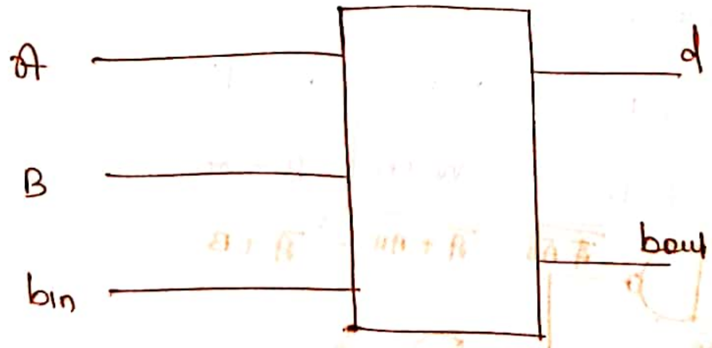
A * <u>HS using NOR gate</u>



· five NOR gate required for Half Subtractor

* Full Substractor

→ It is a Combinational Circuit used to find difference of three binary bits. It has two o/p namely difference and borrow.



A	B	bin	D	B
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

$d(A, B, b_{in}) = \sum m(1, 2, 4, 7)$

difference $\boxed{d = A \oplus B \oplus b_{in}}$

<u>borrow</u>:

A \ B bin	00	01	11	10
0	(0)	(1) 1	(3) 1	(2) 1
1	(4)	(5)	(7) 1	(6)

$$\boxed{B_{out} = \bar{A}\, b_{in} + \bar{A} B + B\, b_{in}}$$

when $A - B - b_{in}$

IIy $B - A - b_{in}$

$$\boxed{B_{out} = A \bar{B} + A\, b_{in} + \bar{B}\, b_{in}}$$

In form of minterm

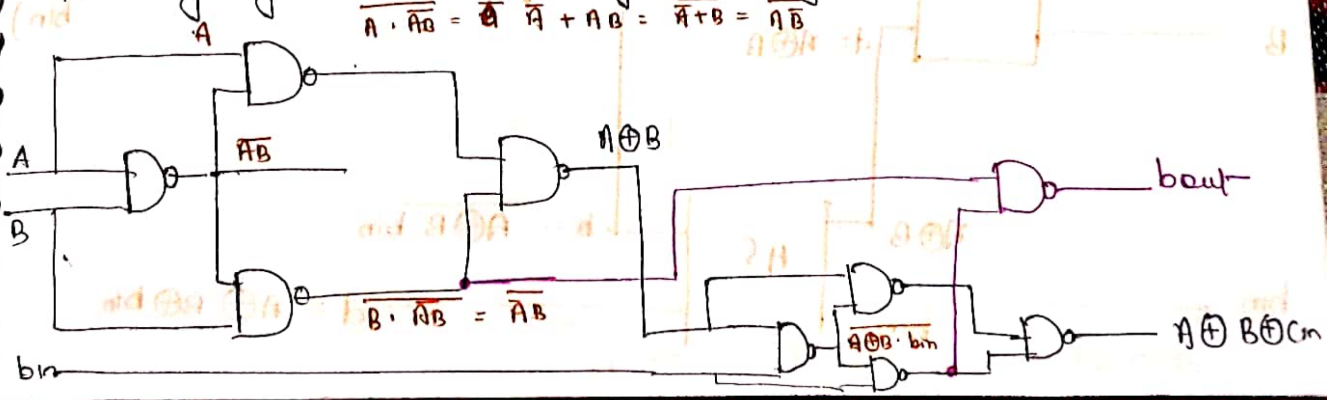
$$b_{out} = \bar{A}\bar{B}\, b_{in} + \bar{A} B \overline{b_{in}} + \bar{A} B\, b_{in} + A B\, b_{in}$$

$$= (\bar{A}\bar{B} + AB)\, b_{in} + (b_{in} + \overline{b_{in}})\, \bar{A} B$$

$$= \overline{A \oplus B}\, b_{in} + \bar{A} B$$

$$\boxed{b_{out} = \bar{A} B + \overline{A \oplus B}\, b_{in}}$$

* Design fullsubtractor using NAND gate

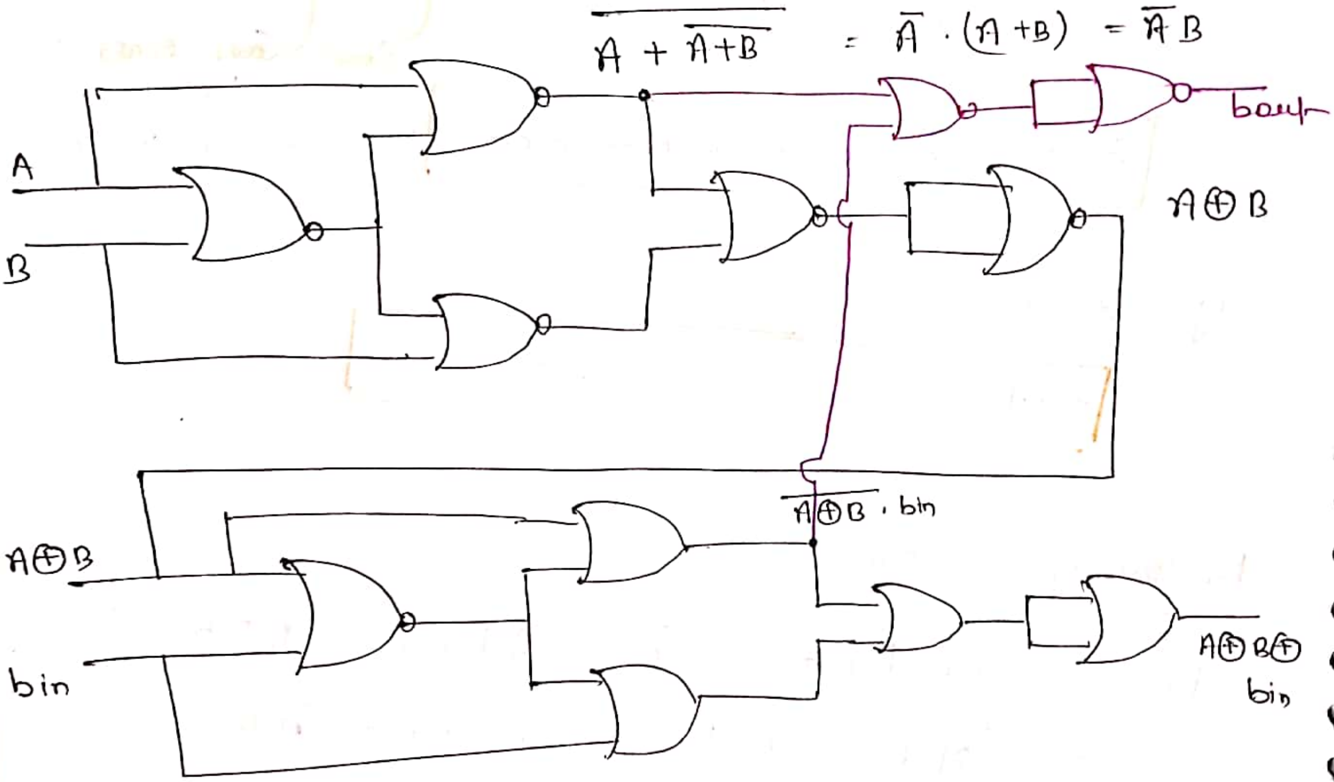


$$\text{borrow} = \overline{\overline{\overline{A}B + (\overline{A \oplus B})\,bin}}$$

$$= \overline{\overline{\overline{A}B} \cdot \overline{\overline{A \oplus B} \cdot bin}}$$

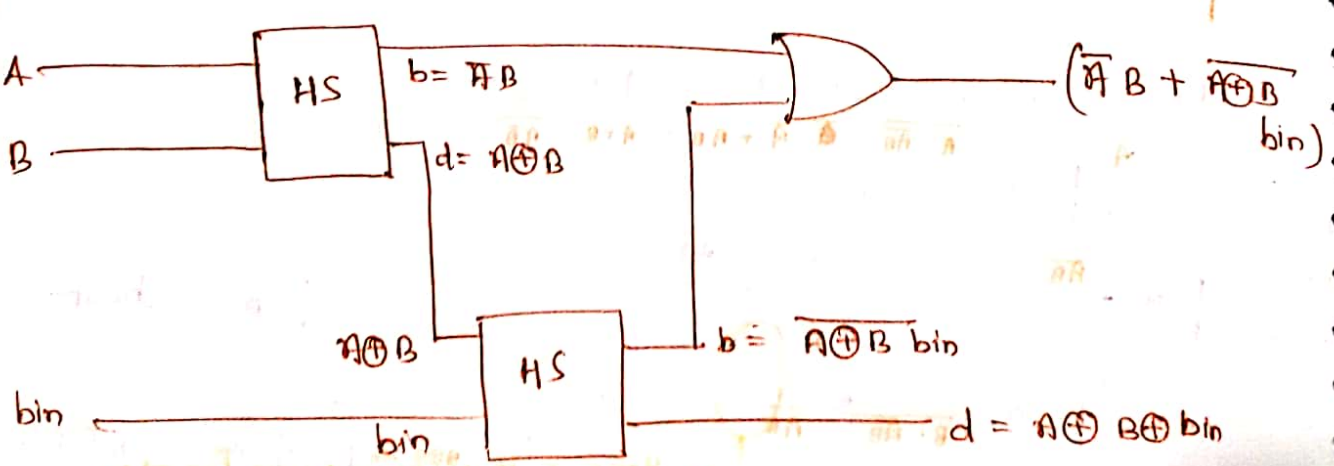
9 nand gates Required

* Design full Subtractor using NOR gate :-



* 12 NOR gates required for designing Full Subtractor

* Full Subtractor using Half Subtractor (2 HS & 1 OR gate)



* Binary Parallel Adder

for addition of two n bit numbers

$$\begin{array}{rcccc} & 0 & 1 & 0 & \\ \text{FA} & 1 & 0 & 1 & 0 \\ & 1 & 0 & 1 & 1 \\ \hline 1 & 0 & 1 & 0 & 1 \end{array} \rightarrow \text{HA}$$

can be done using HA

$$\begin{array}{ccccc} & Cin_3 & Cin_2 & Cin_1 & \\ & A_3 & A_2 & A_1 & A_0 \\ & B_3 & B_2 & B_1 & B_0 \\ \hline Cout_3 & S_3 & S_2 & S_1 & S_0 \\ & & Cout_2 & Cout_1 & Cout_0 \end{array}$$

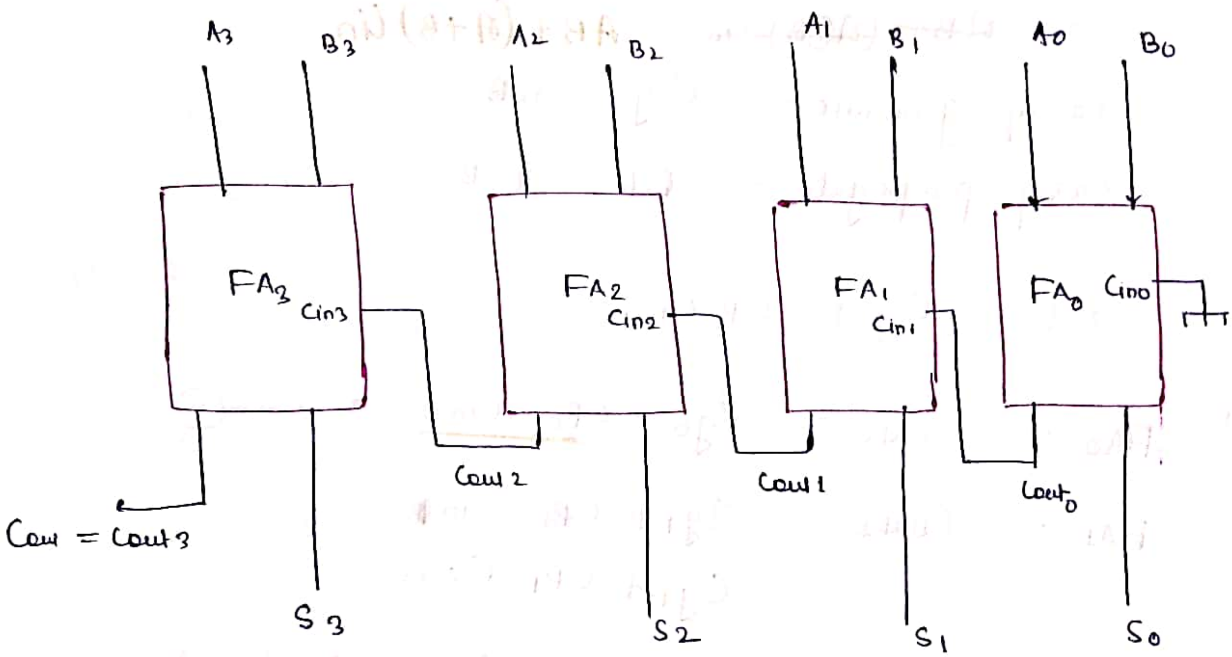
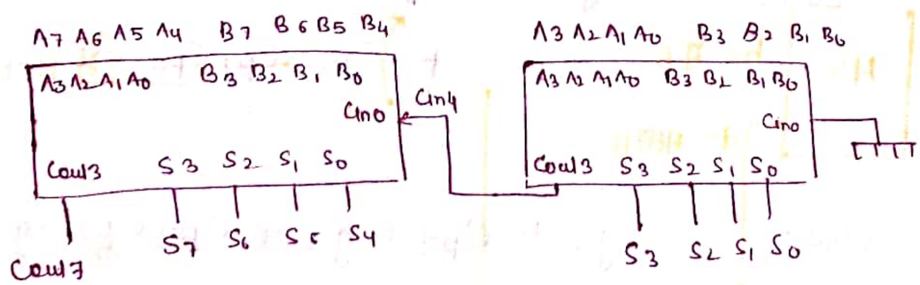


fig. Circuit of Binary parallel Adder

* addition of 8 bit number



$$\begin{array}{cccccccc} A_7 & A_6 & A_5 & A_4 & A_3 & A_2 & A_1 & A_0 \\ B_7 & B_6 & B_5 & B_4 & B_3 & B_2 & B_1 & B_0 \\ \hline \end{array}$$

Here it is necessary to use full adder

- Speed of binary parallel adder is very slow and delay increases with no. of bits. (Disadvantage)
- design is Simple (advantage)

To solve speed problem of Binary parallel adder Look ahead Carry adder is used

* Look A-head Carry Adder

$C_{out} = AB + BC_{in} + AC_{in}$

$= \cancel{AB + (A \oplus B) C_{in}} \quad AB + (A+B) C_{in}$

Carry generate = $C \cdot g = AB$

Carry propagate = $C \cdot P = A + B$

$C_{out} = C_g + C_P \cdot C_{in}$

FA0 :- $C_{out0} = C_{g0} + C_{P0} \cdot C_{in0}$ —— (1)

FA1 :- $C_{out1} = C_{g1} + C_{P1} \cdot C_{in1}$

$= C_{g1} + C_{P1} \cdot C_{out0}$

$= C_{g1} + C_{P1}(C_{g0} + C_{P0} \cdot C_{in0})$

$= C_{g1} + C_{P1} C_{g0} + C_{P1} C_{P0} C_{in0}$ —— (2)

FA2 $C_{out2} = C_{g2} + C_{p2} C_{g1} + C_{p2} C_{p1} C_{g0} + C_{p2} C_{p1} C_{p0} C_{in0}$ —— (3)

FA3 $C_{out3} = C_{g3} + C_{p3} C_{g2} + C_{p3} C_{p2} C_{g1} + C_{p3} C_{p2} C_{p1} C_{g0} + C_{p3} C_{p2} C_{p1} C_{p0} C_{in0}$ —— (4)

Note:- Due to Systematic arrangement C_{out3} cannot take much time. Only two input gates are sufficient to obtain C_{out} in any level.

Que. A 1-bit FA takes 20ns to generate Carry out and 40ns to generate Sum bit. What is the Maximum rate of addition for second when four-bit adder are Cascaded?

A) 10^7 B) 6.25×10^6 C) 1.25×10^7 D) 10^5

Sol $C_{out} = 20ns$

$S = 40ns$

for 1 bit addition

$S = A_0 + B_0 + C_{in0}$ 40 ns

$S = A_1 + B_1 + C_{in1}$ $\underset{C_{out0}}{20} + 40 = 60$

$S = A_2 + B_2 + C_{in2}$ $60 + 20 = 80$

$S = A_3 + B_3 + C_{in3}$ $80 + 20 = 100$ ns

Hence For complete Sum 100ns req.

Rate of add^n $= \dfrac{1}{100 \times 10^{-9}} = 10^7$ / sec

for 16-bit adder

~~$S_{out} = T_{so} + (A-1)$~~

$$\boxed{S = T_{so} + (n-1) T_{cout}}$$

$= 40 + (16-1) 20$

$= 40 + 15 \times 20$

Rate = 1/340ns //.

$S = 340$ ns //.

Last Addition will start at : Cout delay $\times (n-1)$

Last result will be obtained at :- Cout delay $\times (n-1)$ + max [Cout delay , Sum delay]

<u>Que.</u> Given two three bit no. $a_2 a_1 a_0$ & $b_2 b_1 b_0$ and C_1 Carry in. The function that represents Carry generate function when these two numbers are added is

~~A)~~ ~~a_2~~

A) $a_2 b_2 + a_2 a_1 b_1 + a_2 a_1 a_0 b_0 + a_2 a_0 b_1 b_0$
$+ a_1 b_2 b_1 + a_1 a_0 b_2 b_0 + a_0 b_2 b_1 b_0$

B) $a_2 b_2 + a_2 b_1 b_0 + a_2 a_1 a_0 b_0 + a_1 a_0 b_2 b_1 +$
$a_1 a_0 b_2 + a_1 a_0 b_2 b_0 + a_2 a_0 b_1 b_0$

C) $a_2 + b_2 + (a_2 \oplus b_2) + (a_1 + b_1 + (a_1 \oplus b_1)(a_0 \oplus b_0))$

D) $a_2 b_2 + \overline{a_2} a_1 b_1 + \overline{a_2}\,\overline{a_1} a_0 b_0 + \overline{a_2} a_0 \overline{b_1} b_0$
$+ a_1 \overline{b_2} b_1 + \overline{a_1} a_0 \overline{b_2} b_0 + a_0 \overline{b_2}\,\overline{b_1} b_0$

<u>Sol</u> check o/p Carry

Opt A

$a_2 b_2$

$$\begin{array}{ccc} 1 & x & x \\ 1 & x & x \\ \hline 1 & 0 & \end{array}$$

Carry obtain

a_2, a_1, b_1

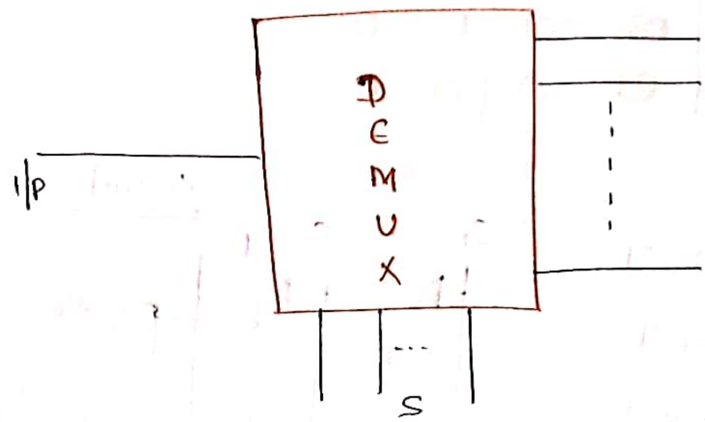
$$\begin{array}{ccc} 1 & 1 & x \\ x & 1 & x \\ \hline 1) & 0 & 0 \end{array}$$

carry obtain

Option B $a_2 b_2 \rightarrow$ generate.

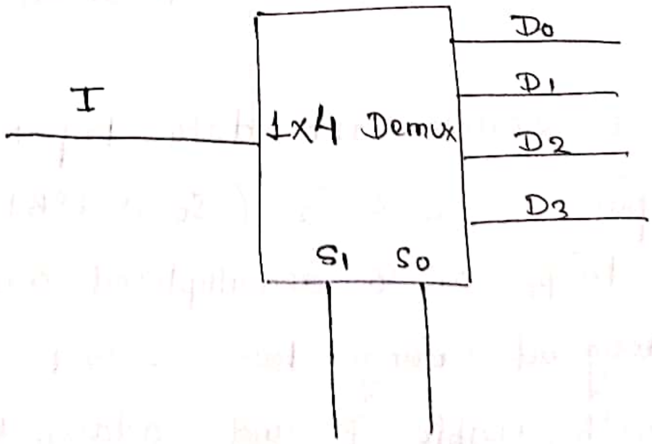
$a_2 b_1 b_2$

* Demultiplexer



- Demux is a Combinational ckt which accepts data from Single Input line and distribute it to Various output (Multiple) ouput line.
- The distribution of data from Single Input line to the Multiple o/p line is done with the help of extra Input line Called as Selection line.

* Block Diagram of 1x4 demux :-



Truth table

S_1	S_0	Output D_0	D_1	D_2	D_3
0	0	I	0	0	0
0	1	0	I	0	0
1	0	0	0	I	0
1	1	0	0	0	I

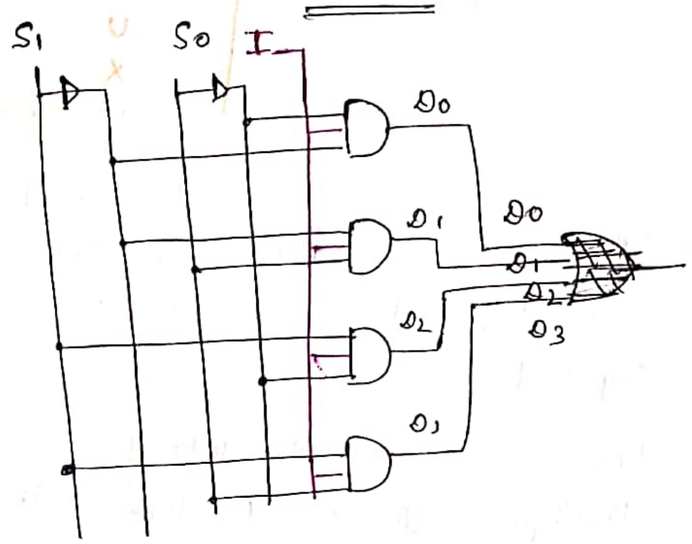
$D_0 = \overline{S_1}\,\overline{S_0}\, I$

$D_1 = \overline{S_1}\, S_0\, I$

$D_2 = S_1\, \overline{S_0}\, I$

$D_3 = S_1\, S_0\, I$

circuit



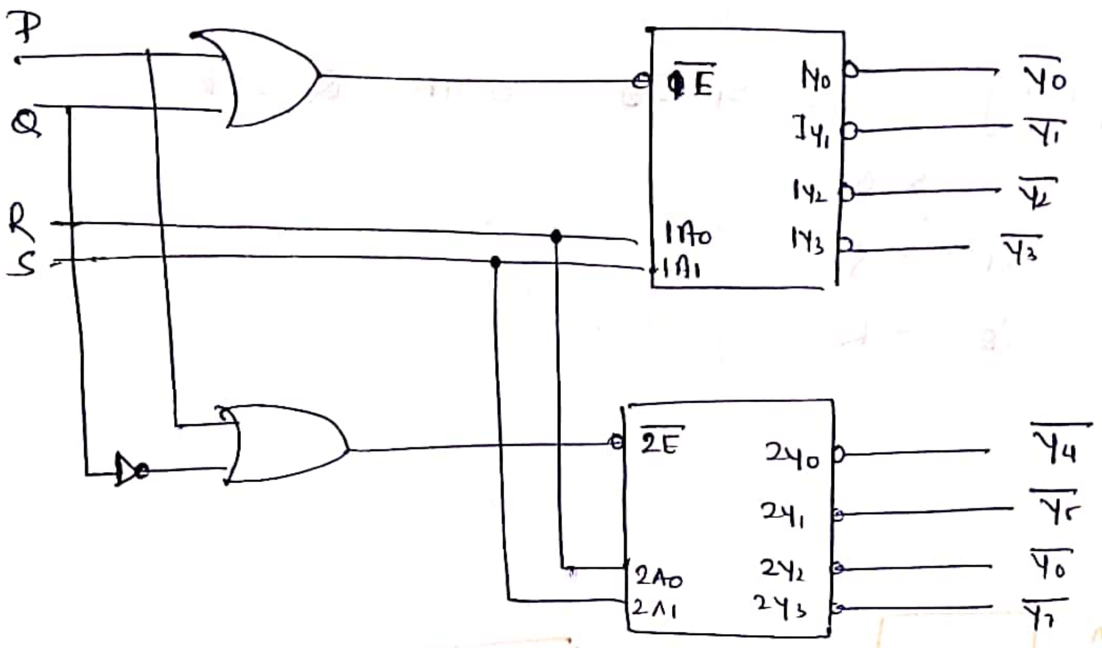
note. MUX :- data selector and parallel to serial converter

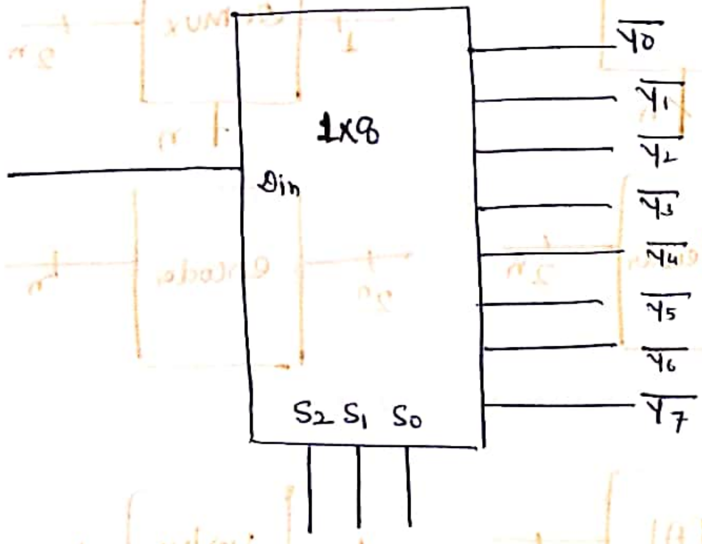
De-MUX :- data distributor and serial to parallel converter.

Que 1 A 1 to 8 Demux with data input D_{in} address input S_0, S_1 & S_2 (S_0 as LSB) and $\overline{Y_0}$ to $\overline{Y_7}$ as 8 demultiplexed outputs, is to be designed using two 2 to 4 Decoders (with enable $\overline{E}$ and address inputs A_0 & A_1) as shown in figure. D_{in} S_0 S_1 & S_2 are to be connected to P Q R S but not necessarily in this order. The respective Input Connection to P Q R & S terminal should be

A) S_2 D_{in} S_0 S_1 B) S_1 D_{in} S_0 S_2

c) D_{in} S_0 S_1 S_2 d) D_{in} S_2 S_0 S_1





Sol: P = 0 fixed (to make device on)

as P = 0 and o/p is active low

hence '0' will propagate from $\overline{Y_0}$ to $\overline{Y_7}$

P = D_{in}

given $A_1 \longrightarrow S$

$A_0 \longrightarrow R$

for $Q = 0$ $\overline{Y_0}$ to $\overline{Y_3}$ activated

for $Q = 1$ $\overline{Y_4}$ to $\overline{Y_7}$ activated

Hence Comparing S_2 S_1 S_0 with Q S R

$S_2 \longrightarrow Q$

$S_1 \longrightarrow S$

$S_0 \longrightarrow R$

Option D

Note: